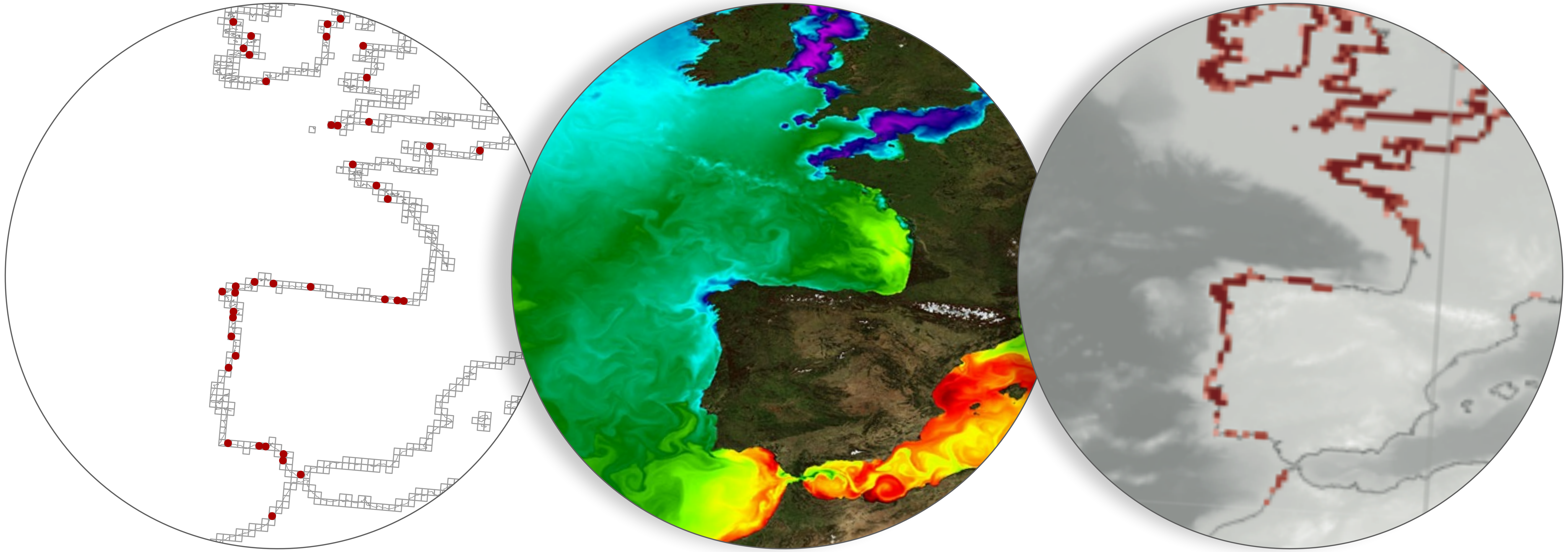
Session 3: The Architecture of Species Distribution Models

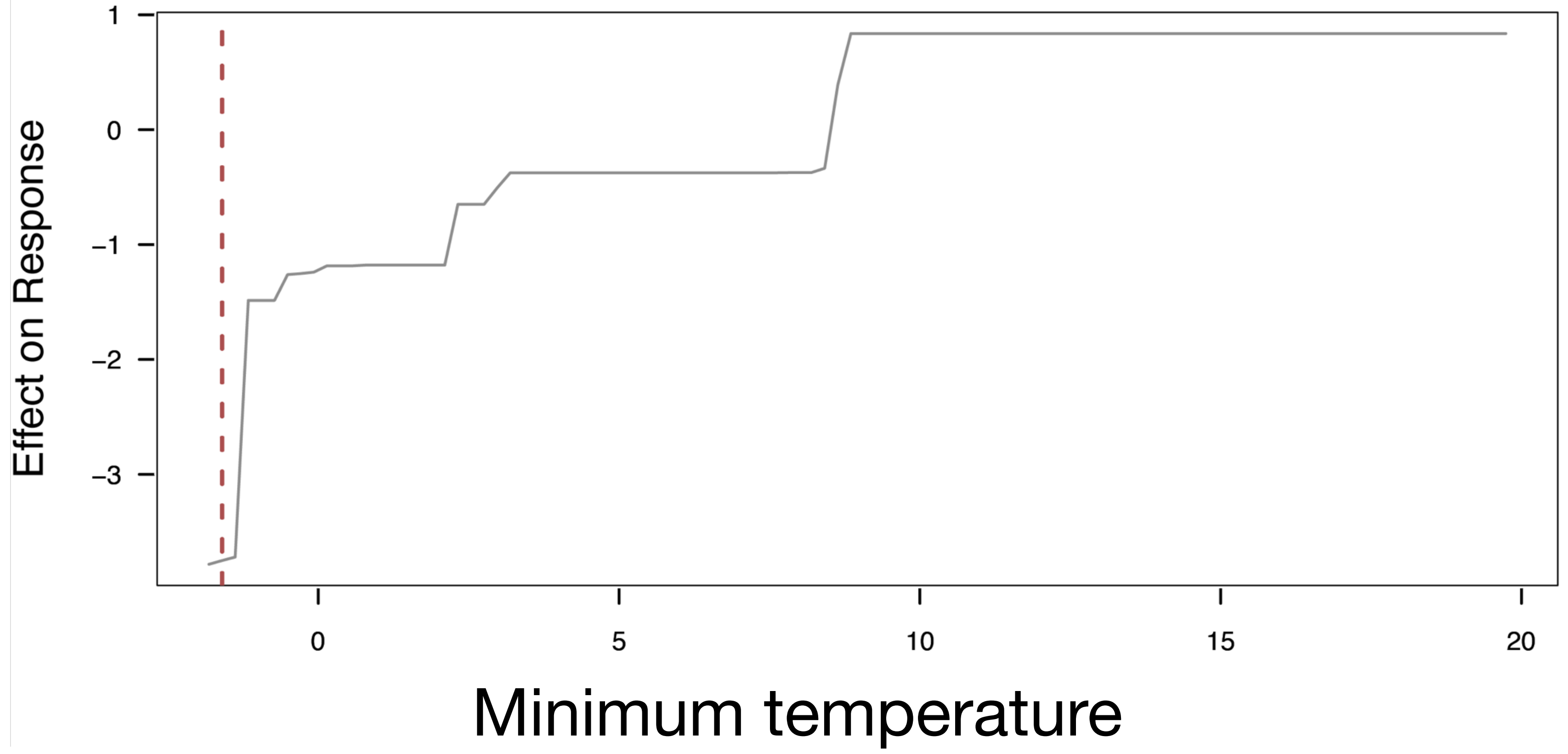
Modeling the distribution of marine biodiversity under global climate change

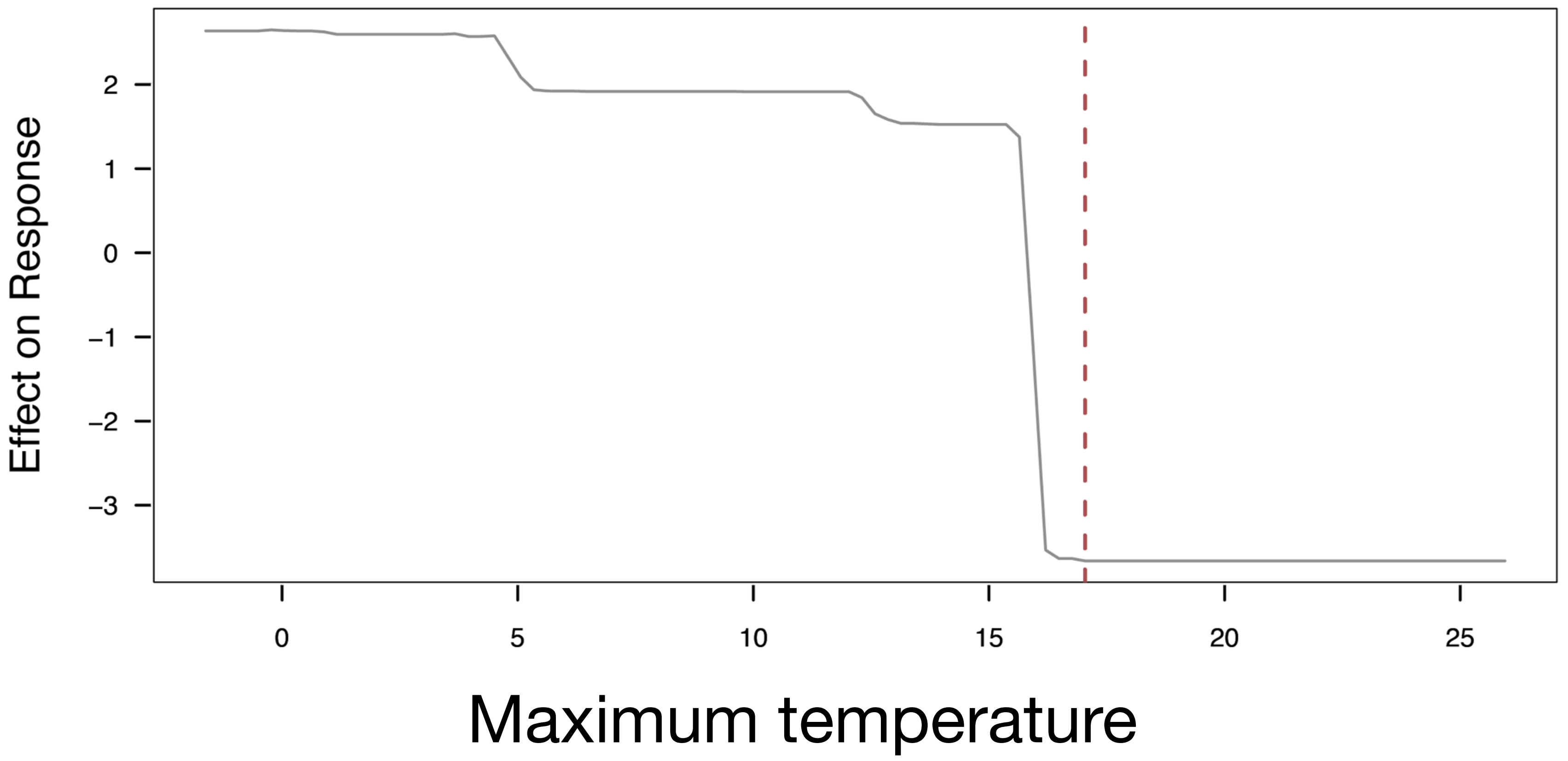
Jorge Assis, PhD // Nord University, Norway // biodiversityDS, Centre of Marine Sciences, Portugal

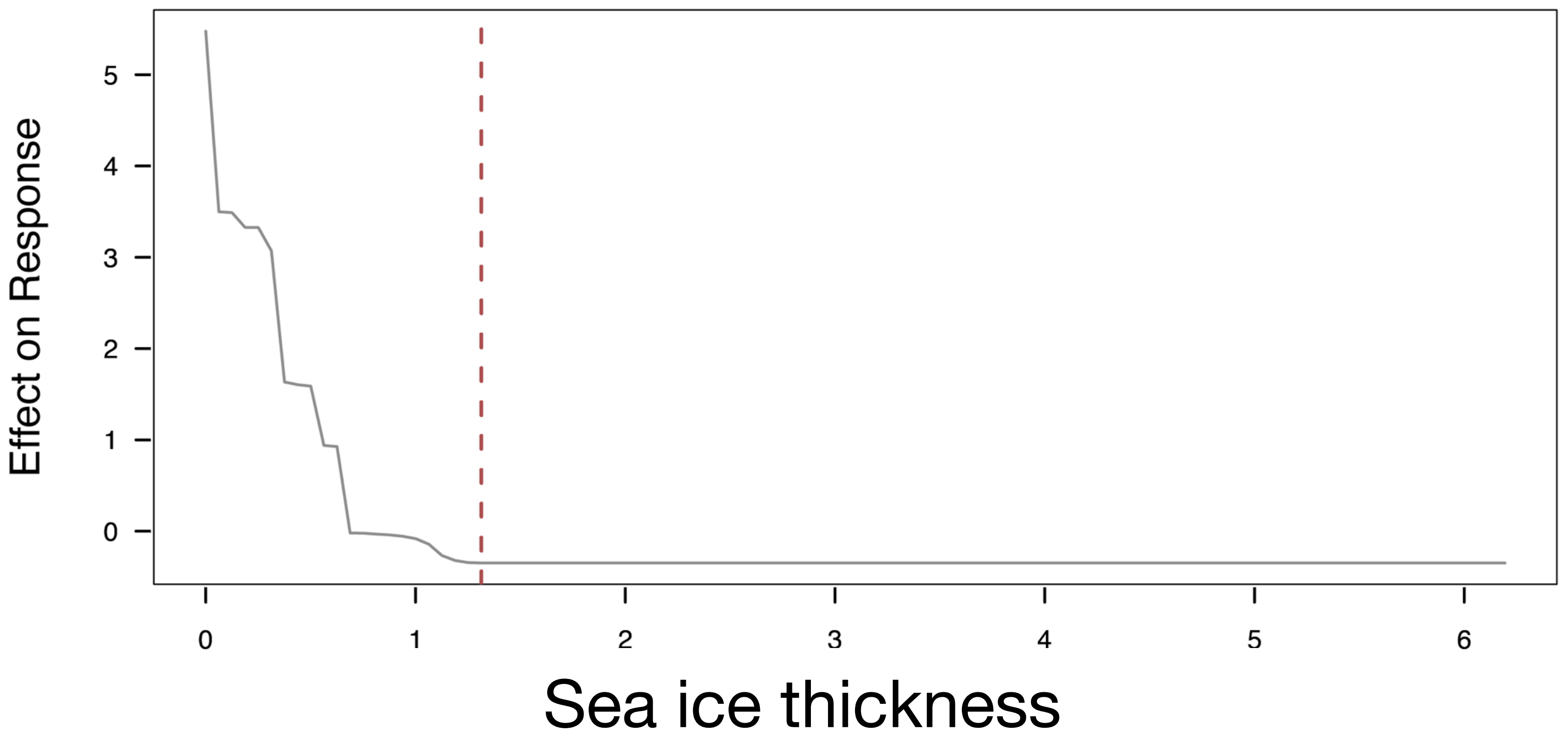
Correlative SDMs

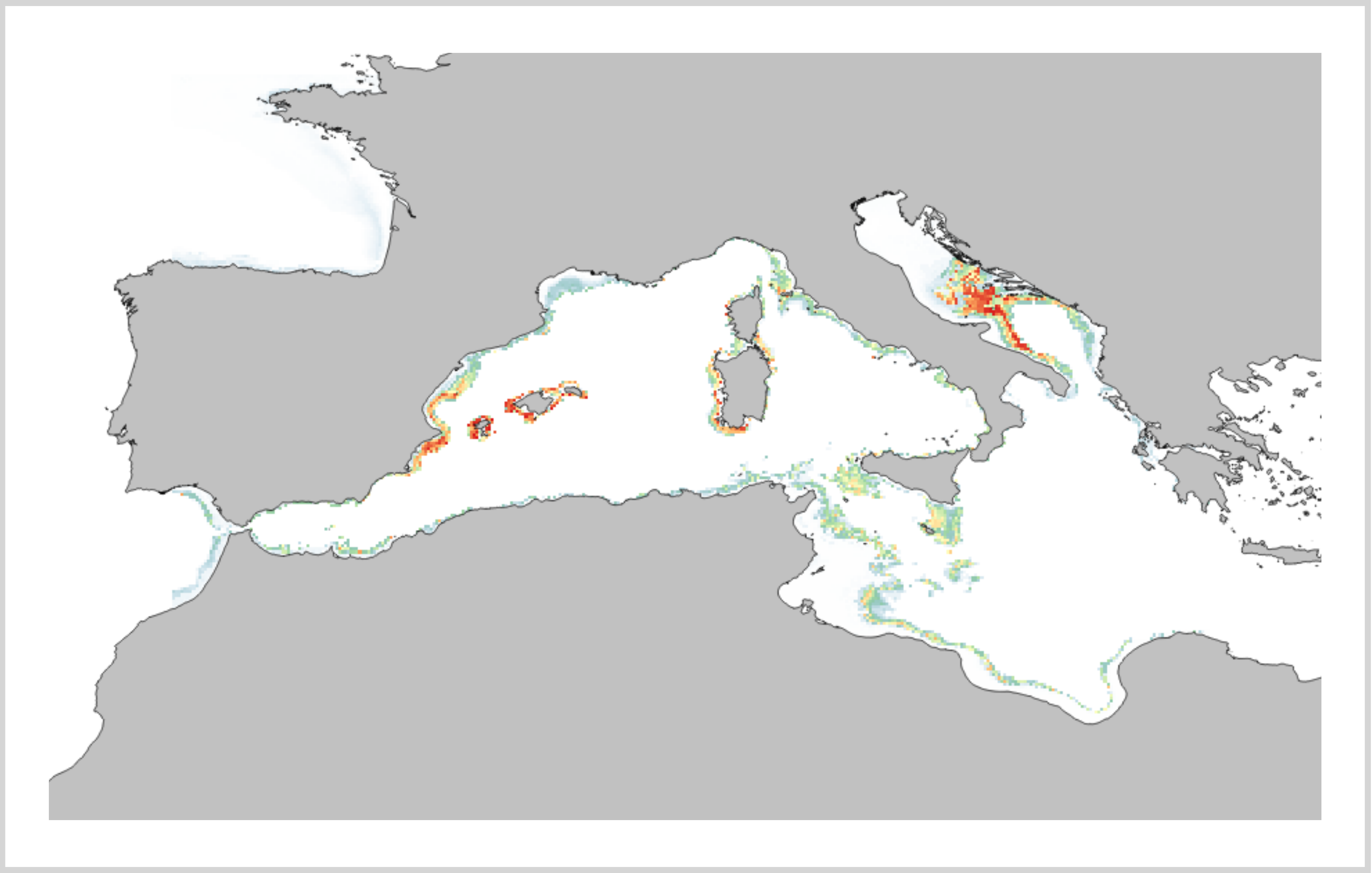


Species distribution modeling using **machine learning algorithms** linking **empirical occurrence records** with corresponding **environmental information** to explain and predict species habitat suitability.

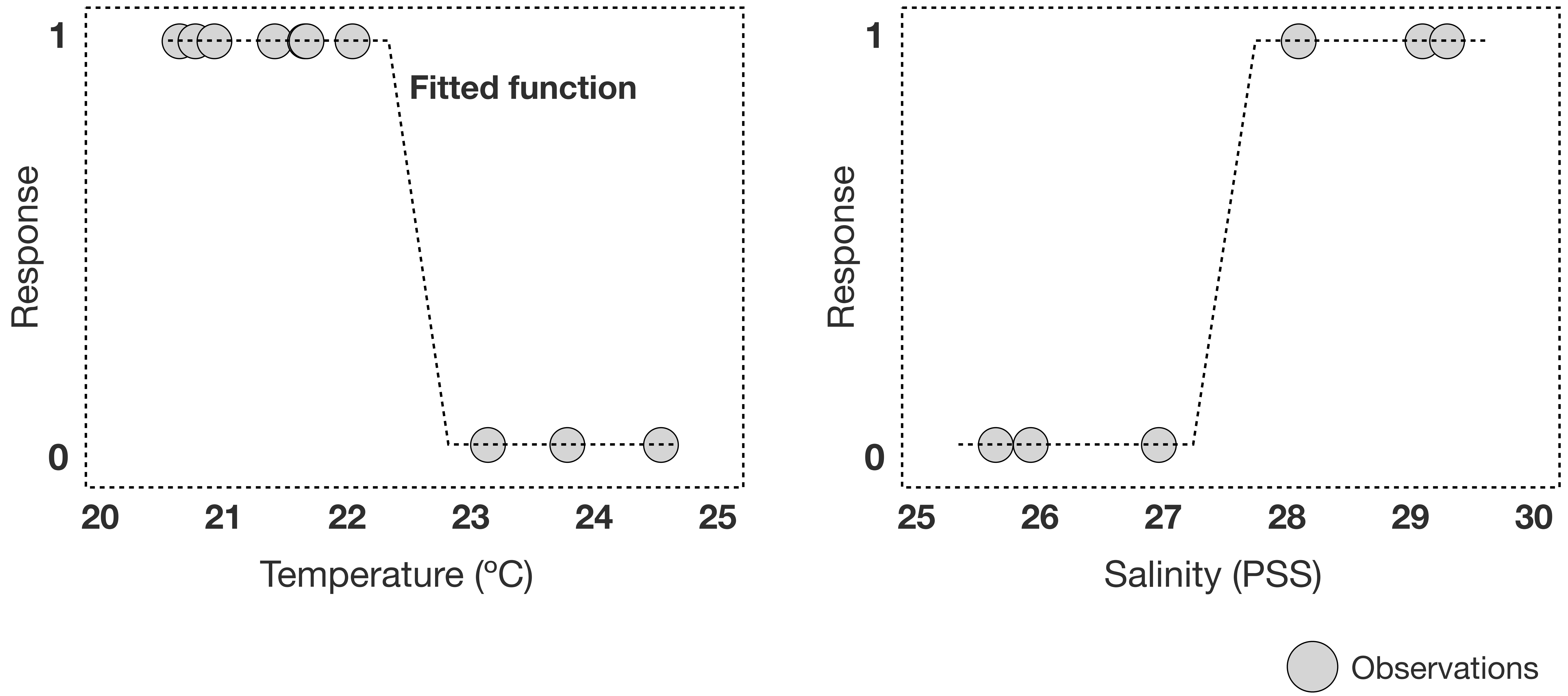






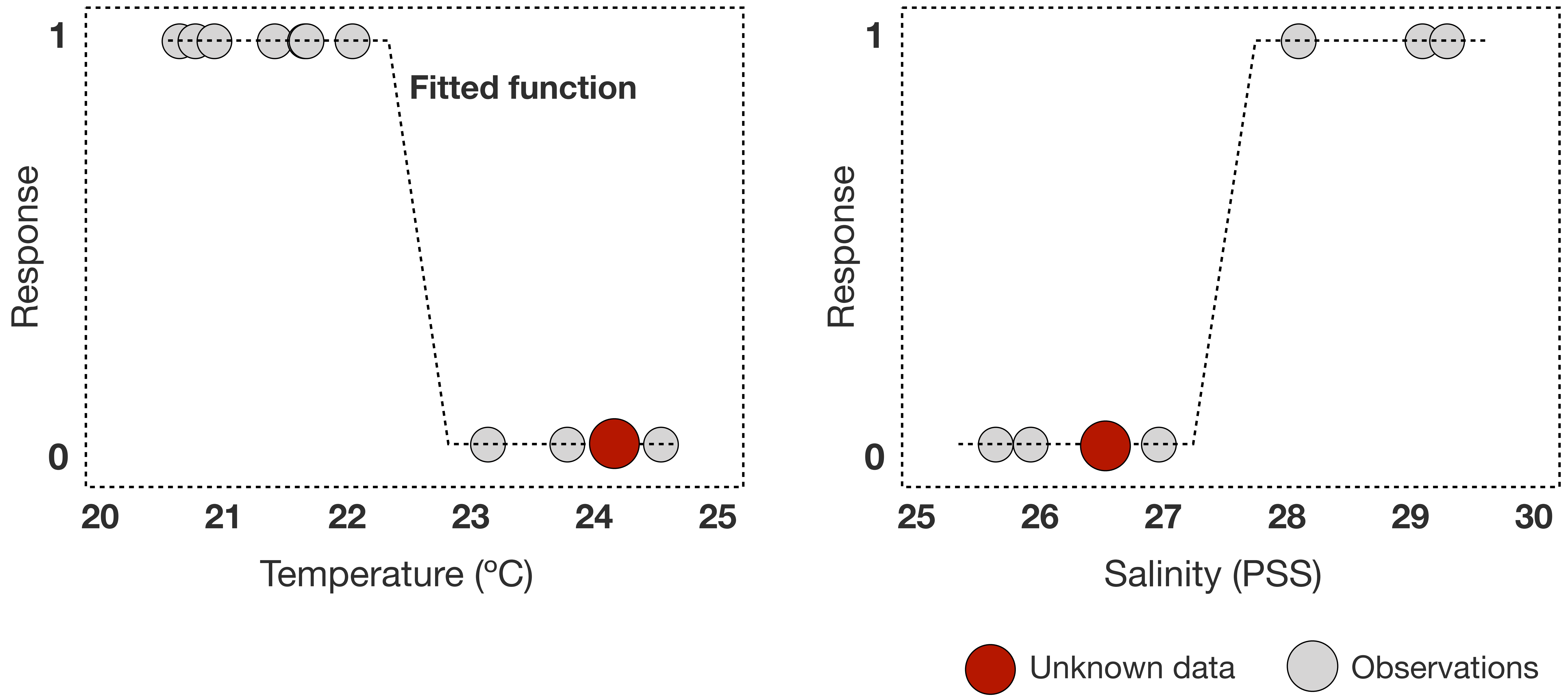


Machine learning (1) captures **the effect of multiple environmental variables** on species distributions; and (2) **predicts species occurrences** as a probability function across space and time (0-1).



Model prediction

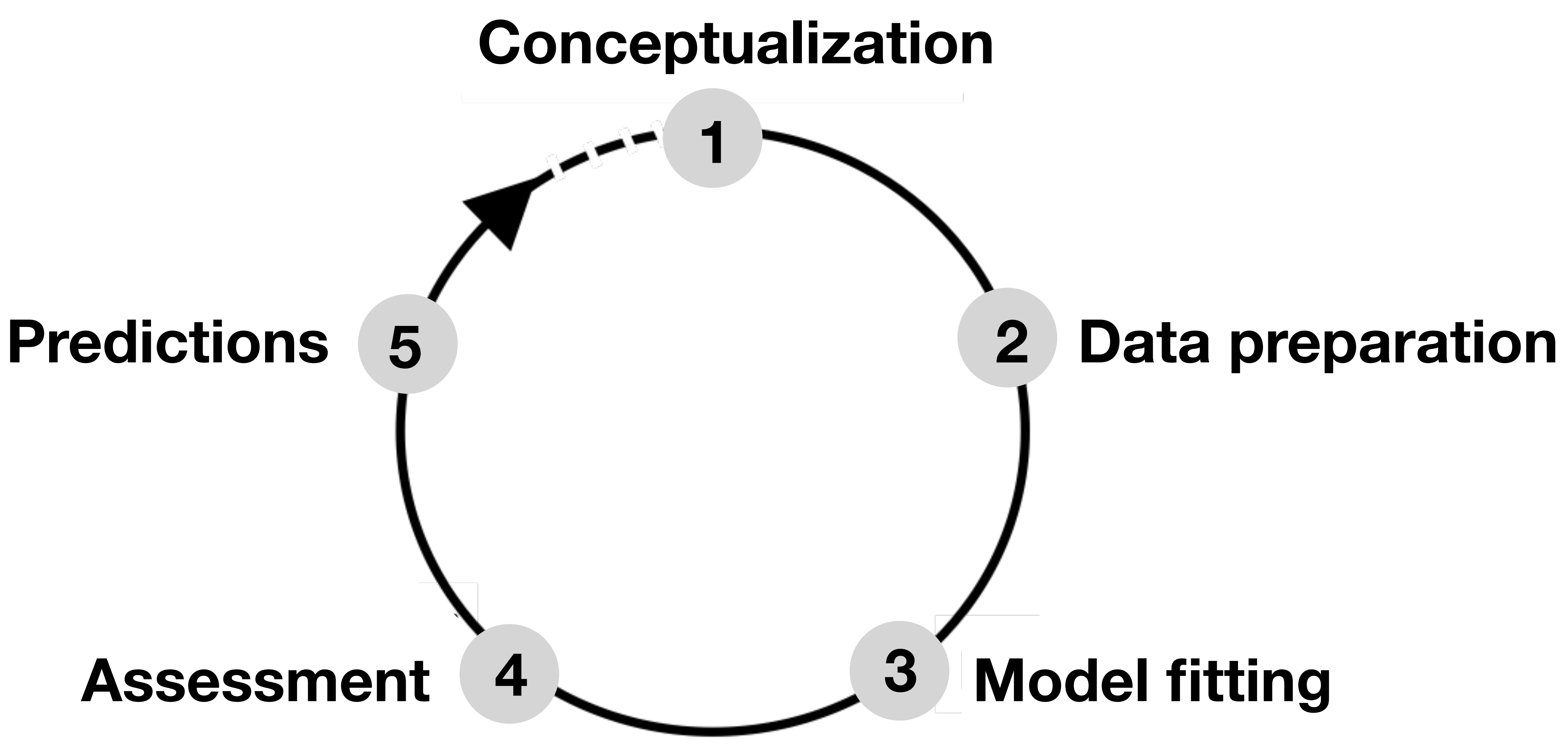
If a model can explain the relationship between distribution records and environmental variables, we can make predictions to unknown samples.



Model prediction

If a model can explain the relationship between distribution records and environmental variables, we can make predictions to unknown samples.

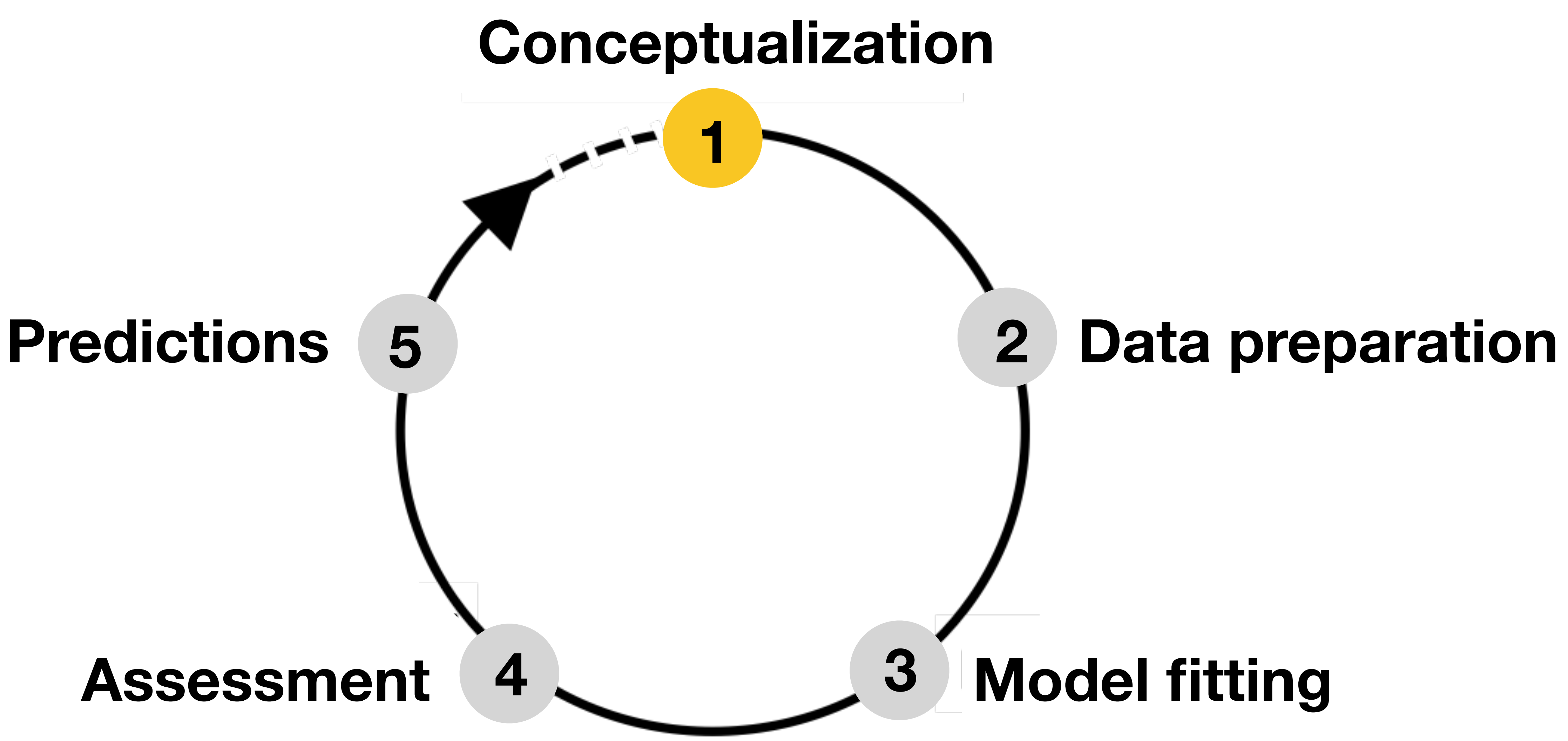
e.g., *Temp. = 24.5ºC & Sal. = 26.5 PSS, response is probability = 0 (i.e., absence).*



Model building: an iterative loop

Feedback loops: building an SDM is not a linear process. It involves distinct steps, **with insights from later steps requiring revisiting earlier ones** (e.g., improve biodiversity data collection or remove surplus environmental layers).

The **goal is to build ecologically meaningful models fit for the purpose** (e.g., conservation planning and climate impact assessment).

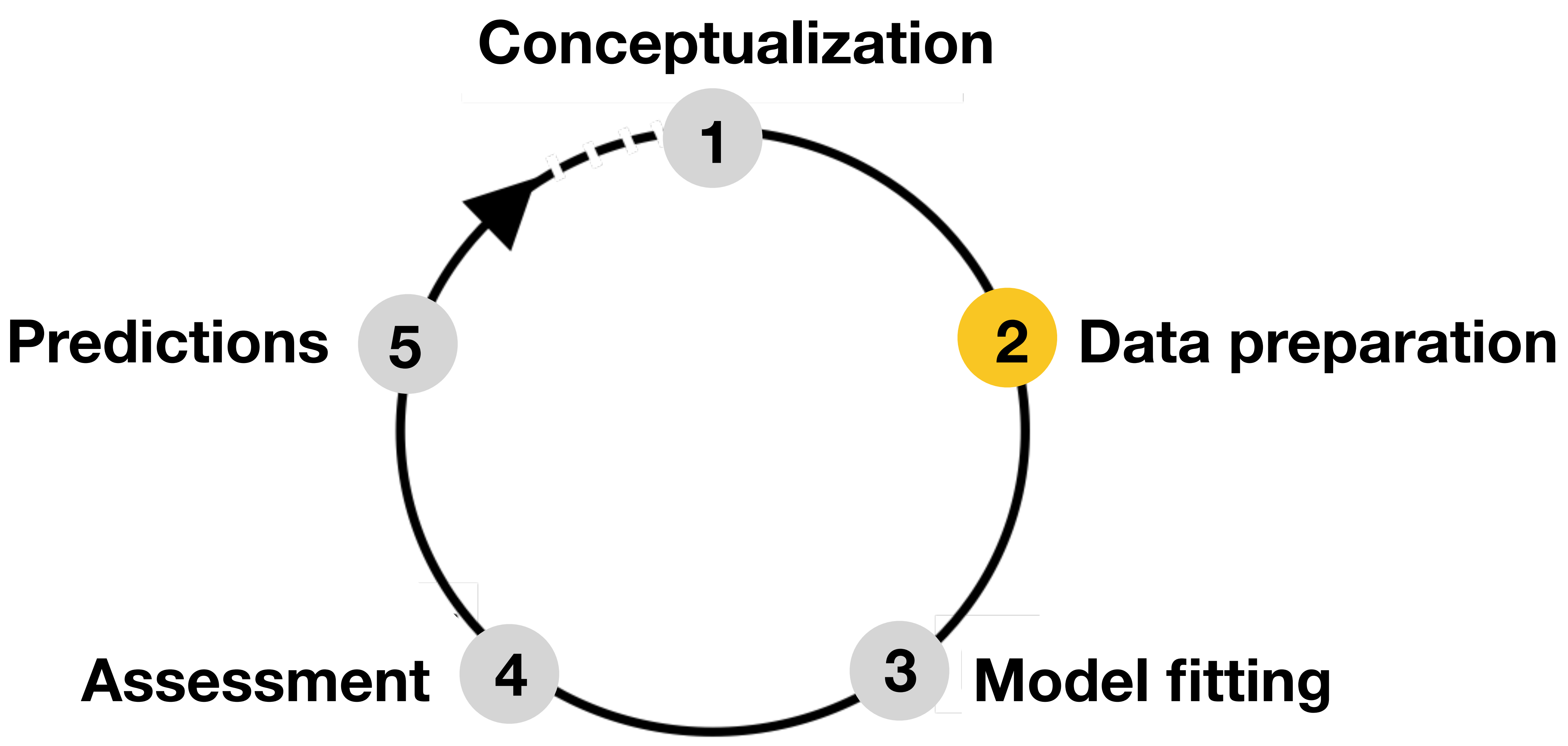


Conceptualization

Define the objective: what question is the model trying to answer?
(e.g., forecast future distribution under climate change and identify invasion risk)

Get knowledge on the species: about its distribution limits (literature).

Hypothesize drivers: What variables may influence its distribution? (literature).

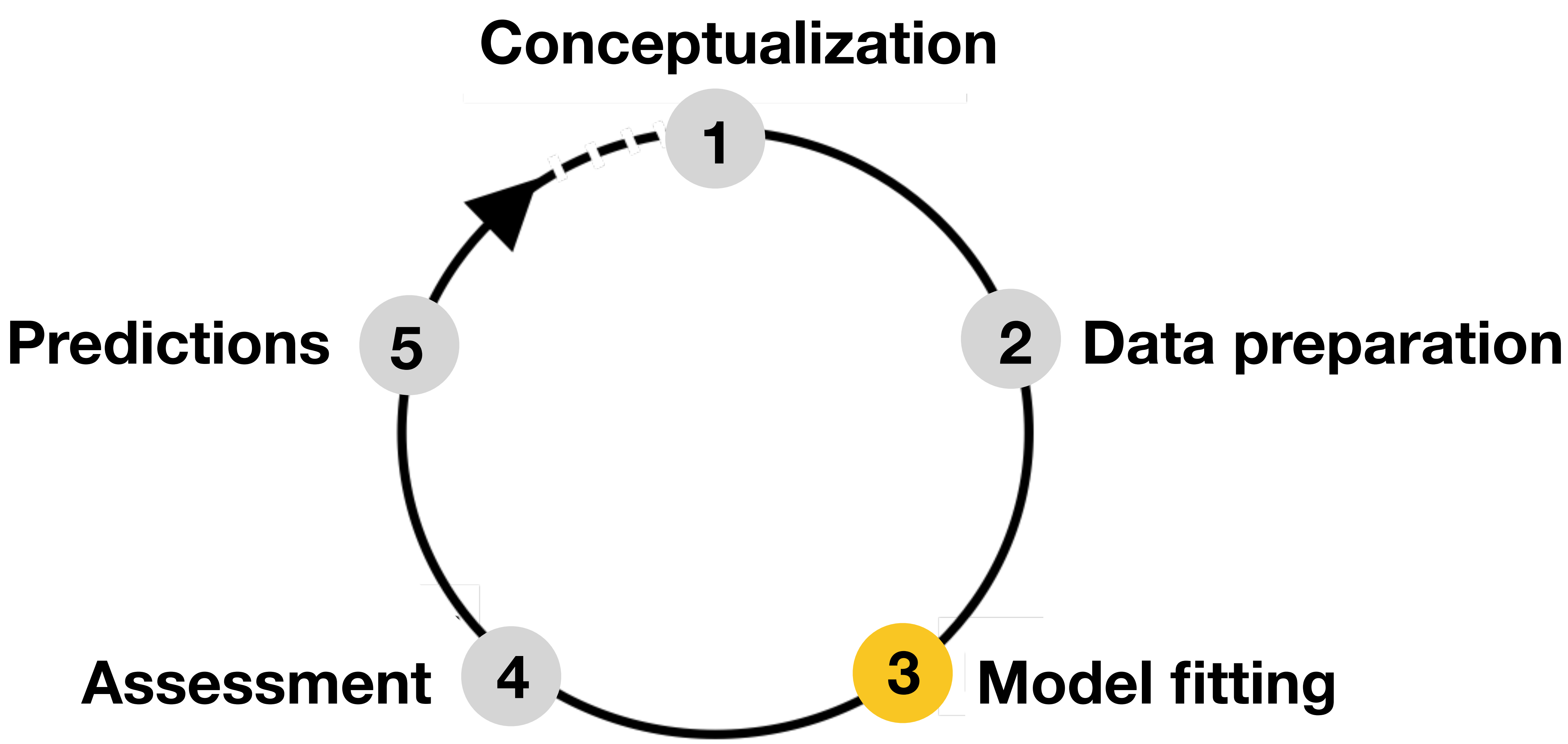


Data preparation

Gather and process species occurrence data (e.g., check for taxonomic errors);

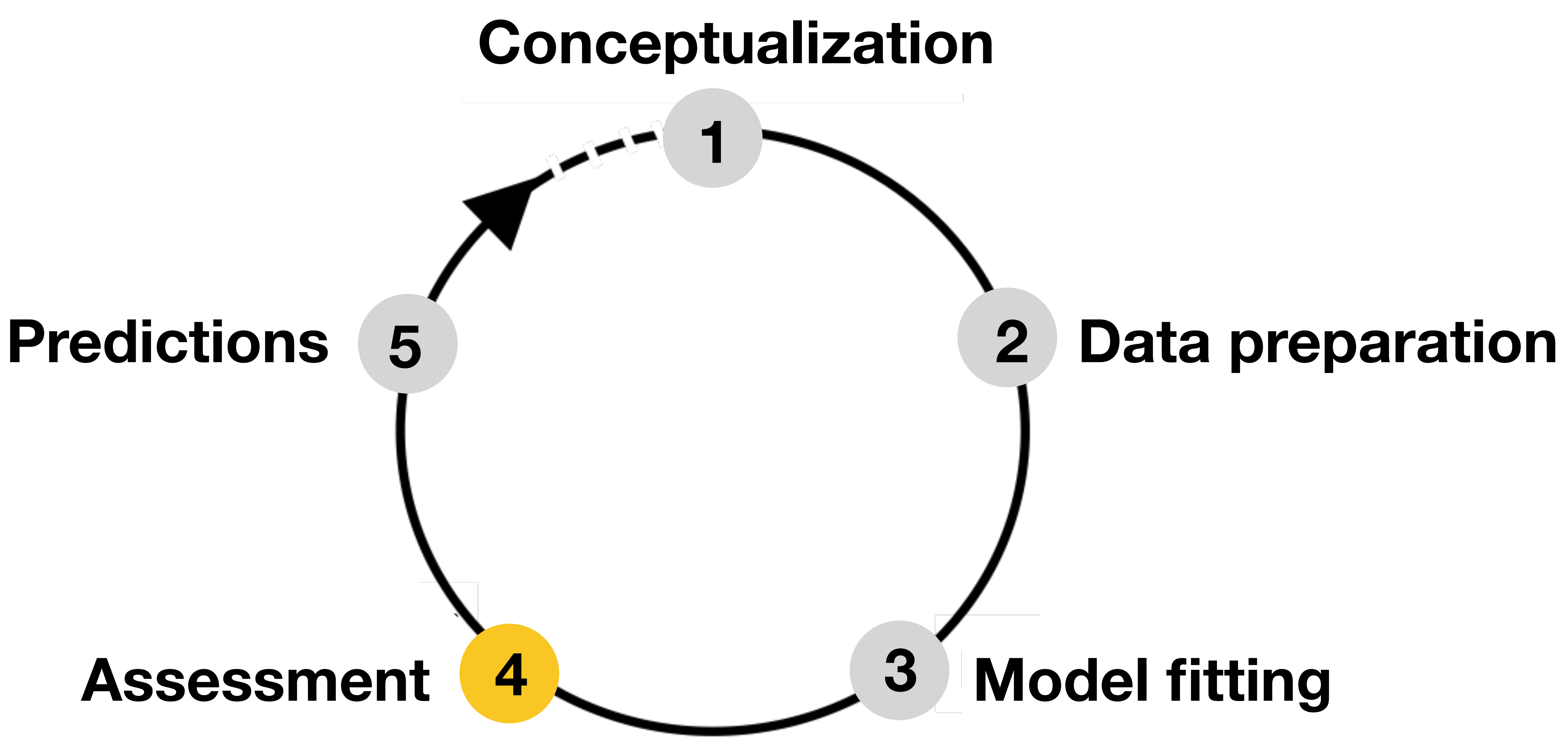
Gather and process environmental variables relevant to the hypotheses.

Define the study area: what geographic extent and spatial / temporal resolution are appropriate for the research question and data?



Model fitting

Algorithm selection: choose a statistical method and the algorithm-specific parameters (e.g., regularization in MaxEnt, tree complexity in BRT).



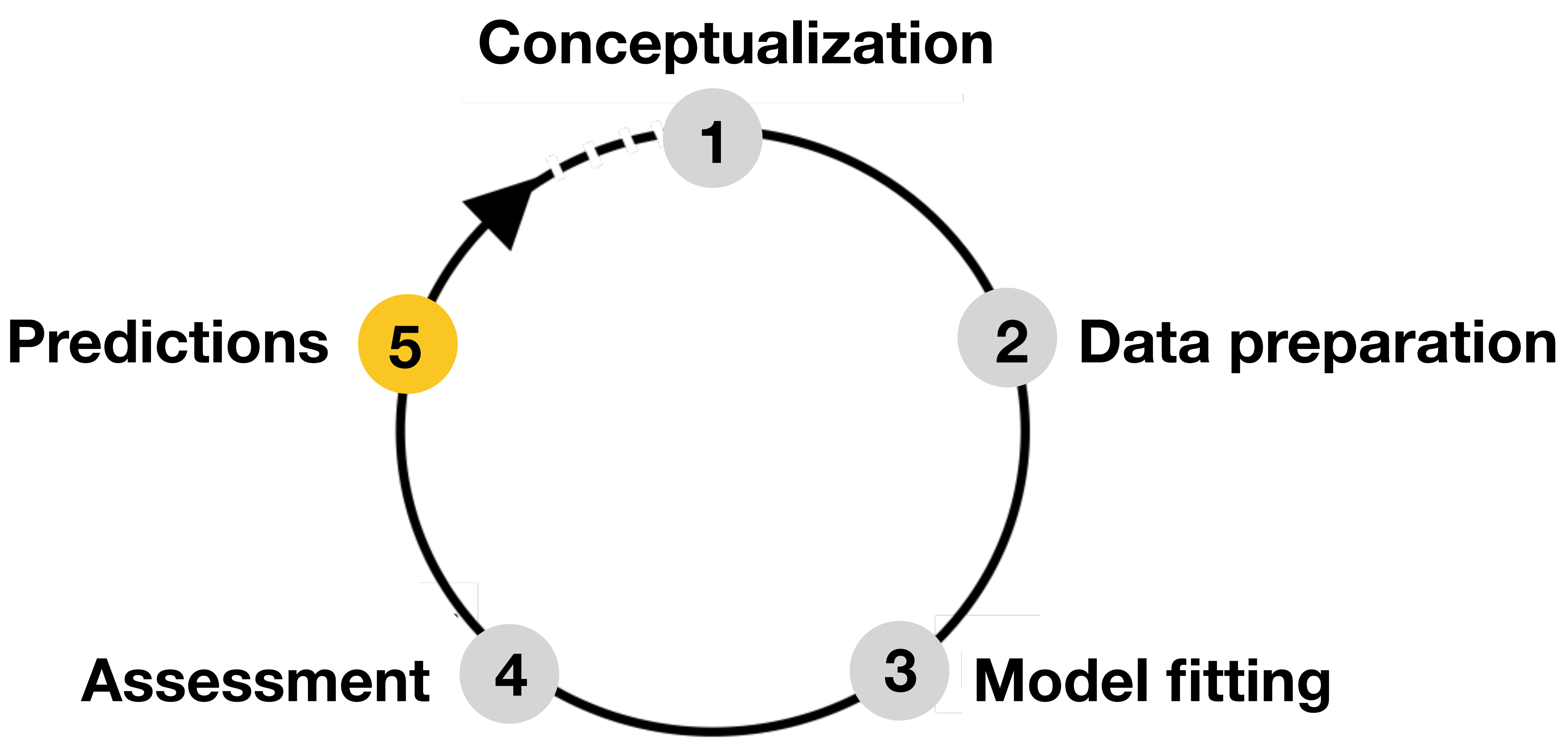
Assessment of the model

How good is the model?

Evaluating performance, as well as ecological realism of model fitting.

Does the model accurately distinguish presence sites?

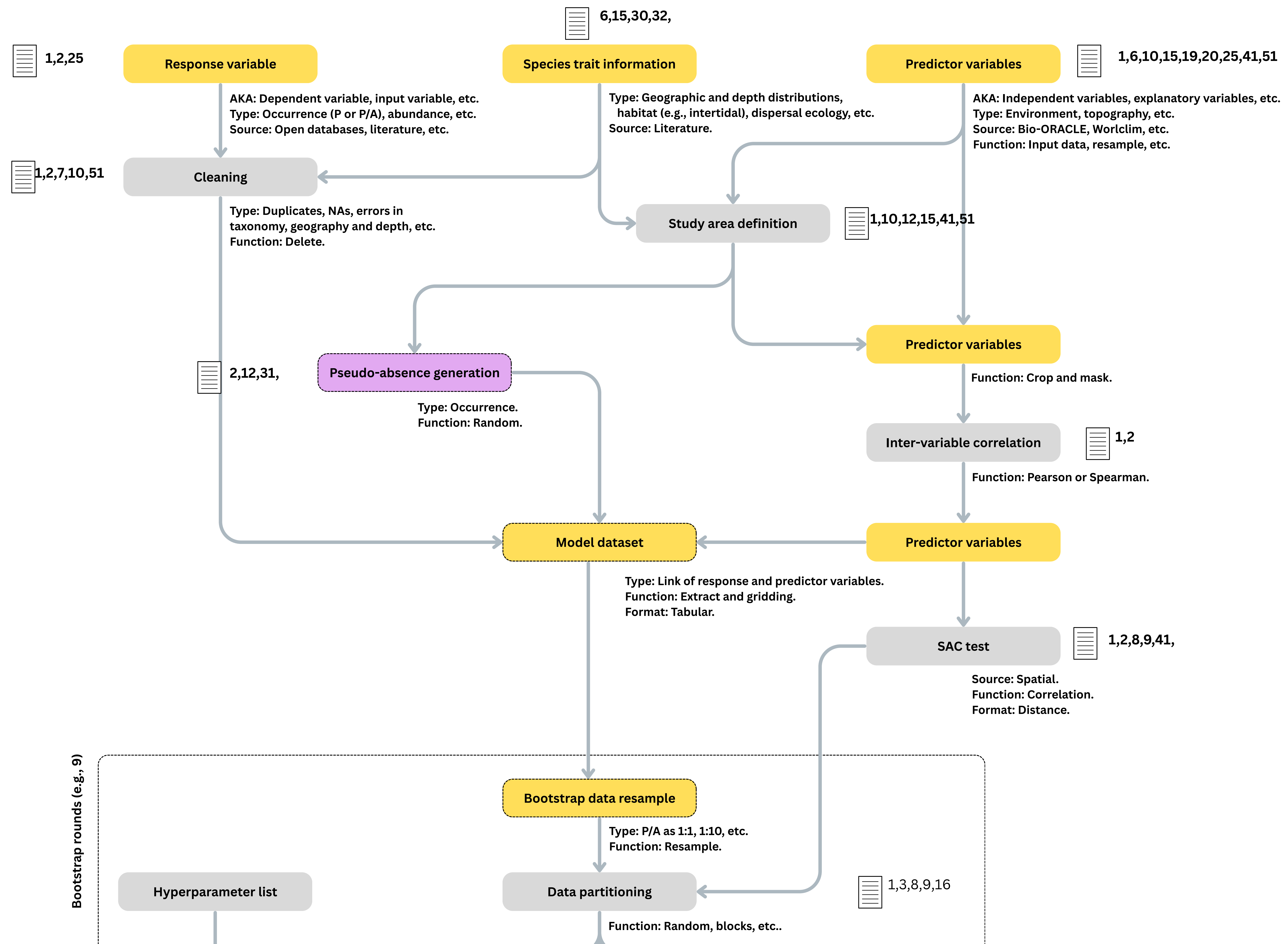
Do the modeled relationships make biological sense? (ecological realism check)



Predictions in space and time

Spatial projection: apply the fitted model to environmental data across the study area to map species' probability of occurrence.

Application: predictions address the initial conceived objective (e.g., map future range shifts, assess invasion risk, identify conservation priorities).


6,15,30,32,
1,2,25
Response variable
Species trait information
Predictor variables
1,6,10,15,19,20,25,41,51
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1,2
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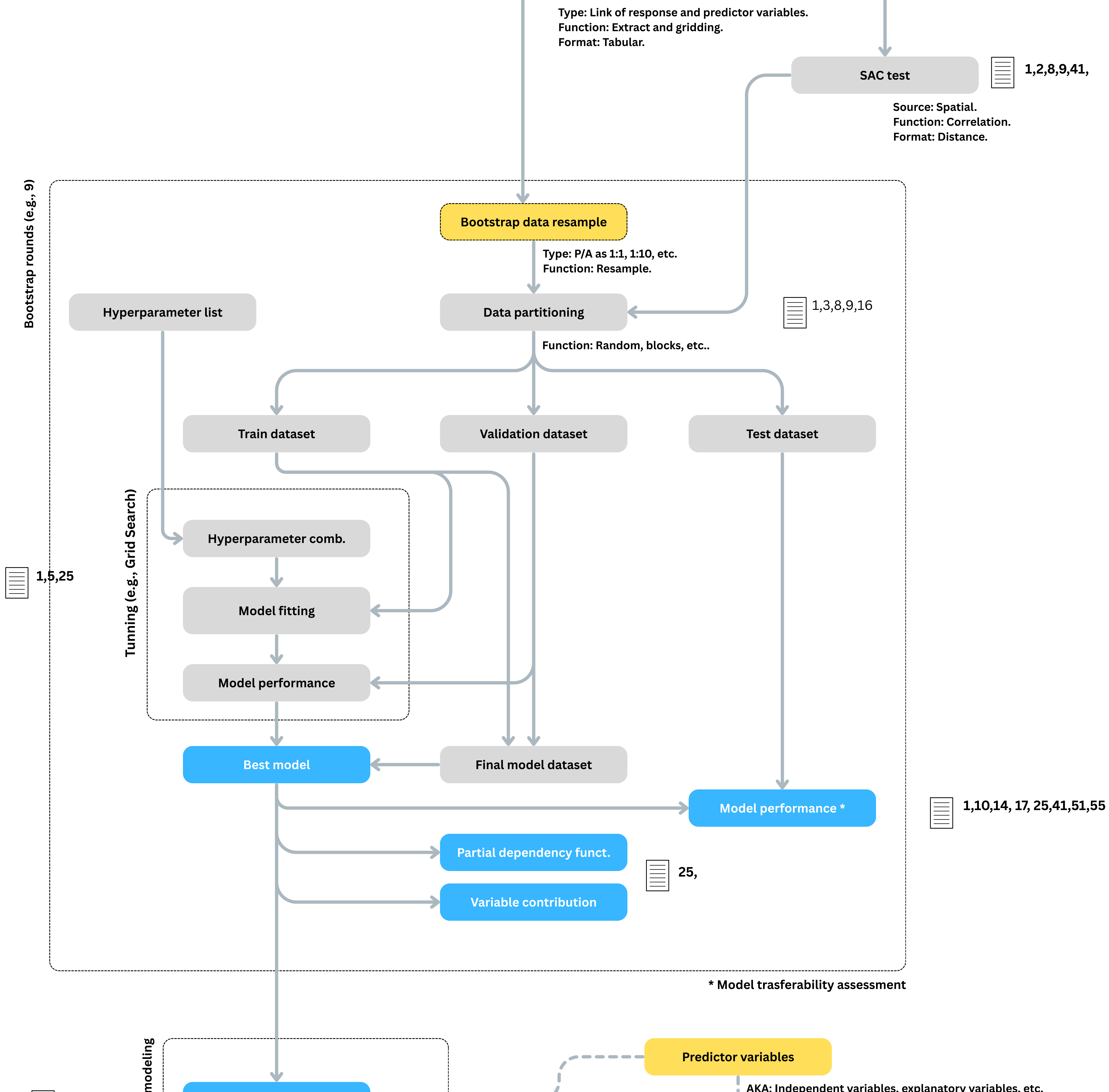
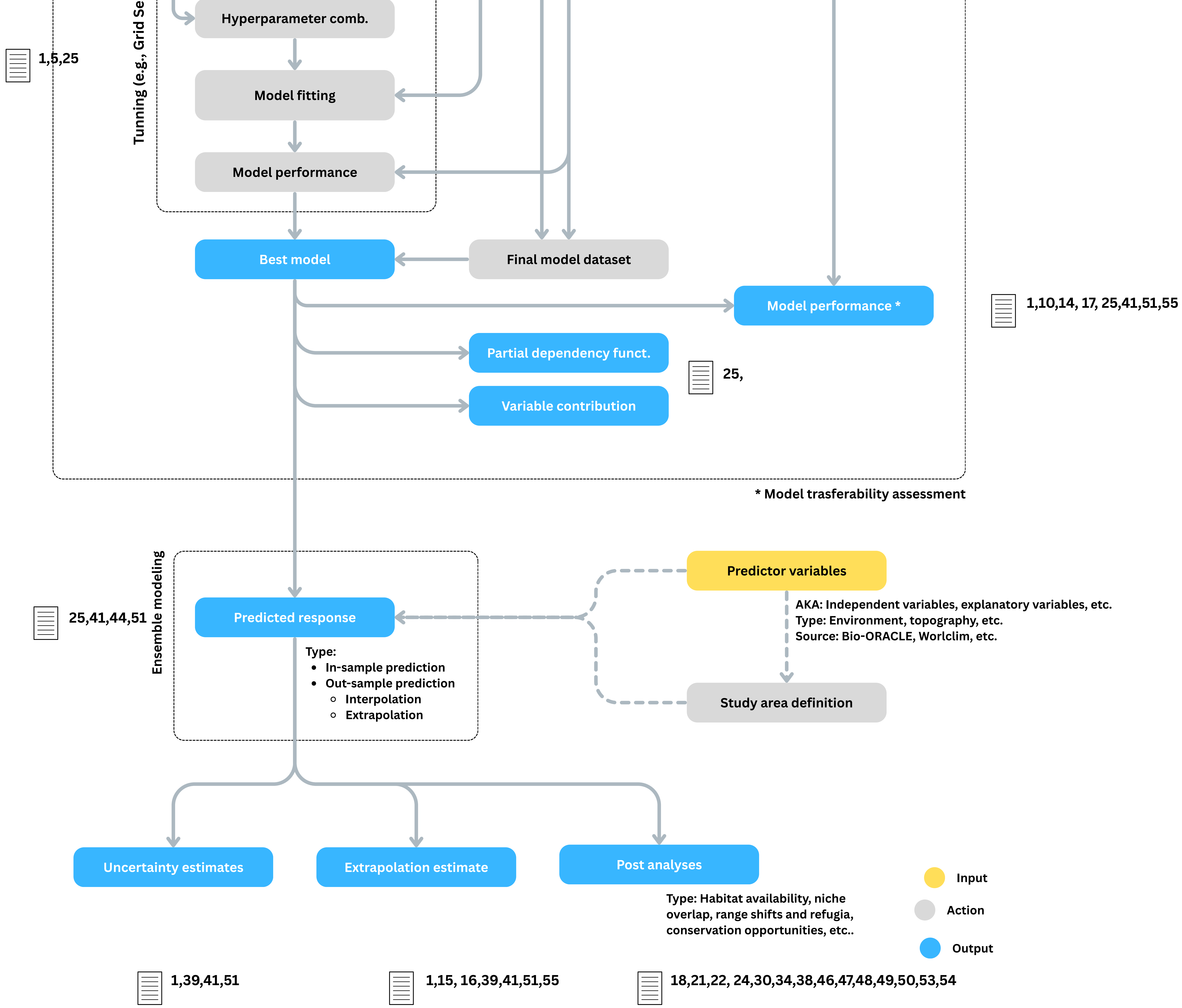
biodiversityDS.





Hyperparameter comb.
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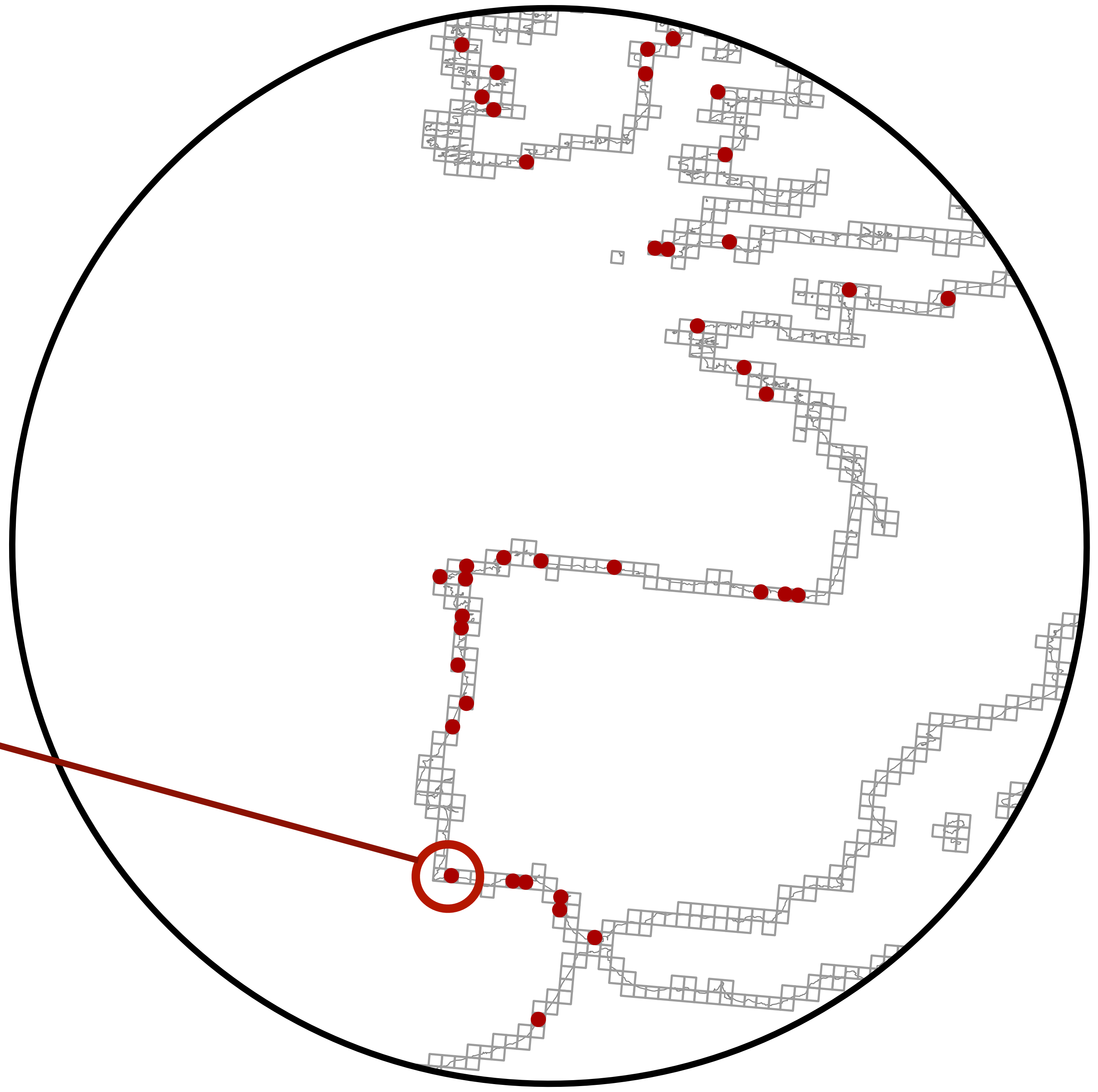
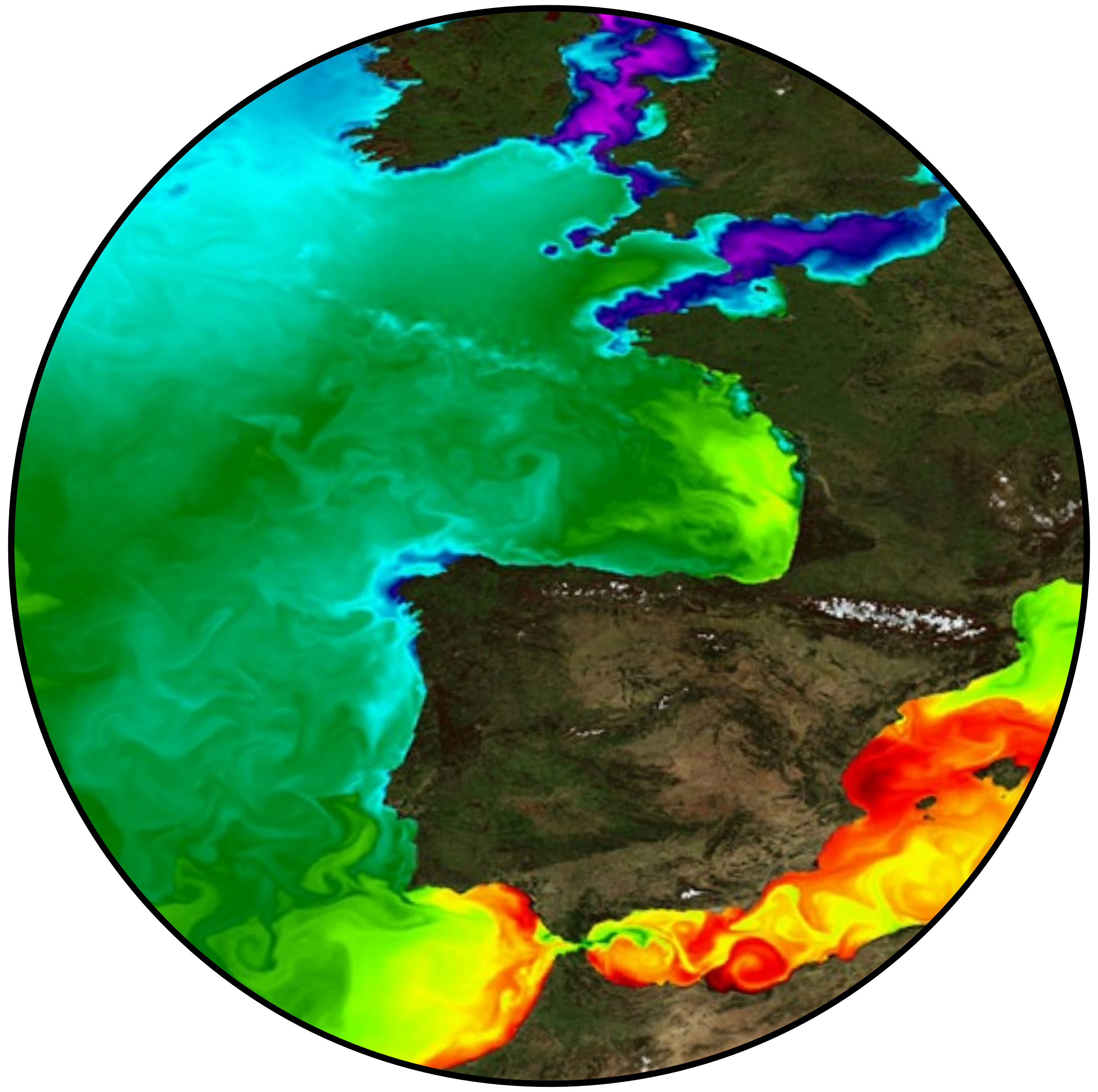
1,5,25
Model fitting
Model performance
Best model
Final model dataset
Model performance *
1,10,14, 17, 25,41,51,55
Partial dependency funct.
25,
Variable contribution
* Model trasferability assessment
Ensemble modeling
Predictor variables
25,41,44,51
Predicted response
AKA: Independent variables, explanatory variables, etc.
Type: Environment, topography, etc.
Source: Bio-ORACLE, Worlclim, etc.
Type:
In-sample prediction
Out-sample prediction
Interpolation
Extrapolation
Study area definition
Uncertainty estimates
Extrapolation estimate
Post analyses
Input
Type: Habitat availability, niche overlap, range shifts and refugia, conservation opportunities, etc..
Action
Output
1,39,41,51
1,15, 16,39,41,51,55
18,21,22, 24,30,34,38,46,47,48,49,50,53,54

Data Preparation

Obs.	SST	Salinity	Oxygen
1	**21.0**	**35.4**	**210.1**
1	19.4	29.1	224.9
1	21.5	31.1	199.8
0	21.3	33.4	201.1
0	14.8	34.1	184.8
(...)	(...)	(...)	(...)



Records of species distribution



Environmental data

Linking Biology to Environment: from Maps to Matrices

Algorithms cannot read maps; they require a data matrices. Achieved by overlaying vector data (**species occurrences**) onto raster grids (**environmental layers**) - extracting the exact pixel values at those coordinates.

"Garbage In, Garbage Out": Spatial Biases

Model outputs are only as good as the input data

The quality of several sources (e.g., GBIF) has been questioned.

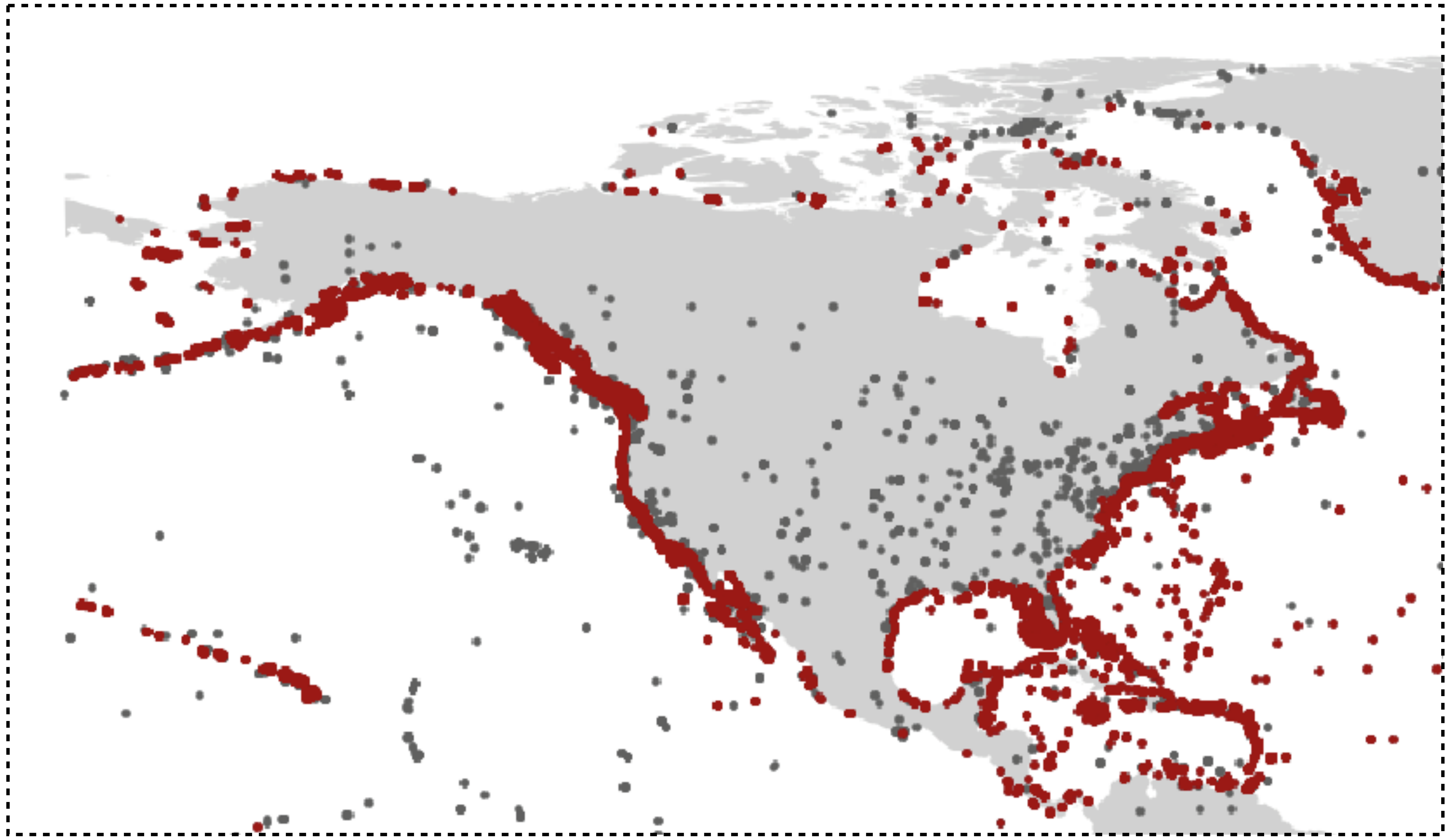
There are **inaccuracies** and **bias** in occurrence records.

Inaccurate records

Inaccurate taxonomy of species

Problem: Records in regions where the species does not exits.

Manual **curation and** expert **judgment is needed** prior to modeling - the molders needs to **know the distribution of the species** to properly judge and curate the occurrence records - the geographic and vertical distribution (e.g., marine realm)



Inaccurate records

Marine records over land (and vice versa)

Problem: Records that are wrong and will not provide relevant information for the model.

Can also arise from coordinate imprecision, flipped Lon-Lat or wrong source information (the museum location instead of records). The modeler needs to discard unreliable coordinate data.

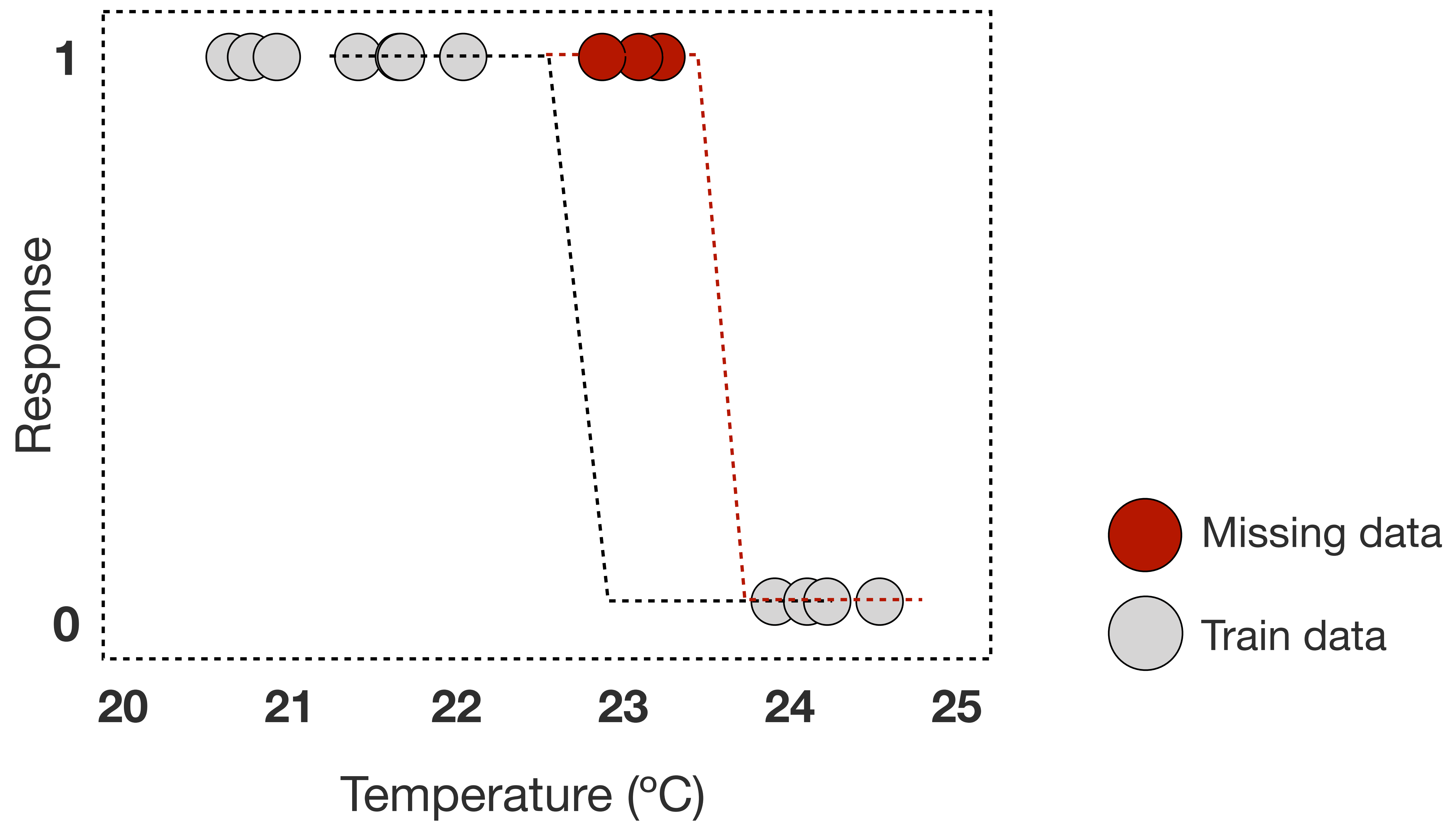
How many presence occurrence records?

Sample size can be positively correlated to the performance of models.

Experiments point to **at least 50 records to estimate response functions, with at least 10 per variable.**

The **number of observations is less important than having well-distributed observations** - cover the environmental space where the species is distributed. Thus, compiling records should be guided by knowledge on species distributions.

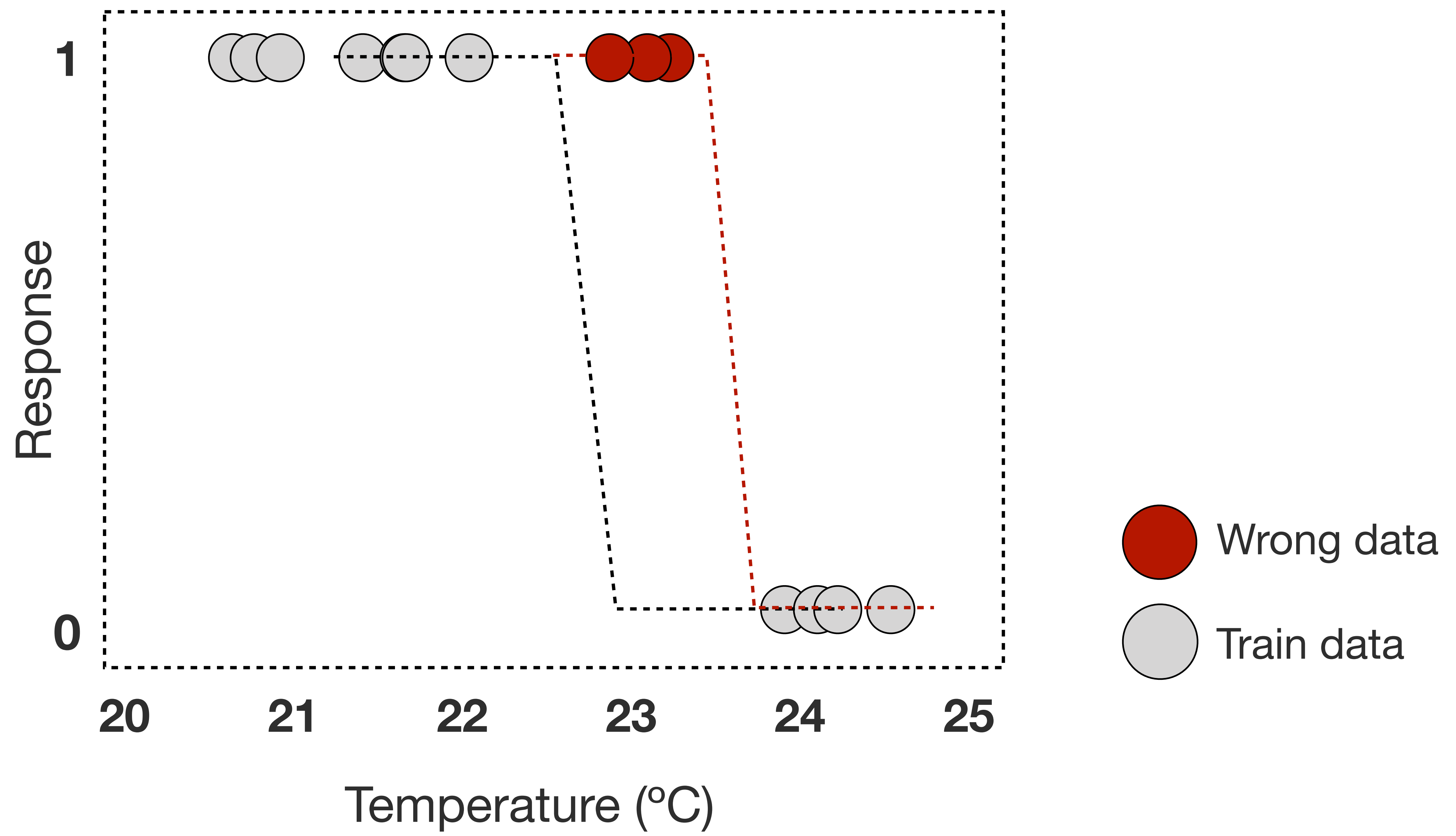
Fei & Yu, 2015



Niche truncation

If presence records are missing, the models may not capture the full width of the niche, i.e., the span of conditions which the specie is exposed to.

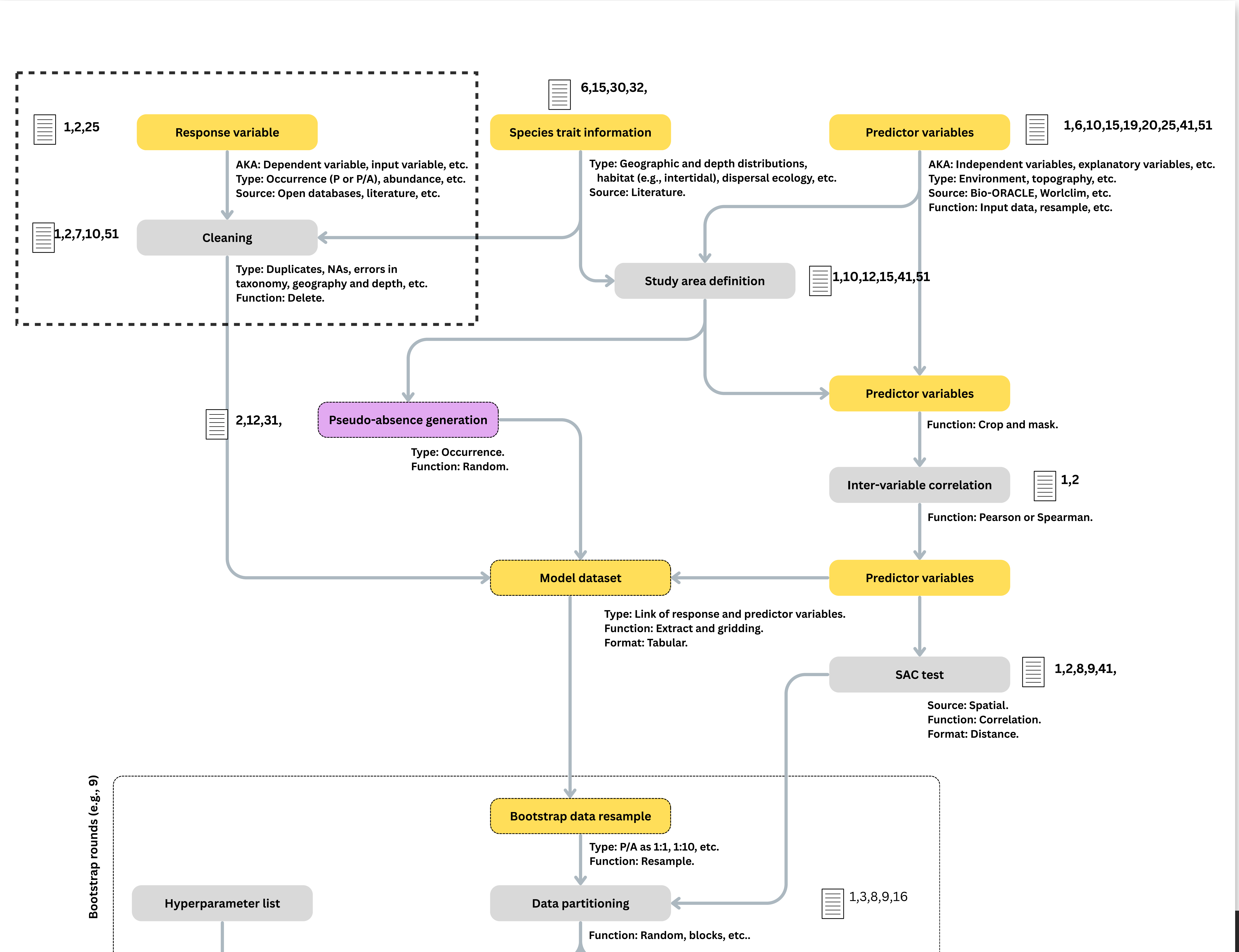
The model fits the portion of the niche represented by the observed records.



Niche expansion

If presence records are wrong, the models may capture a broader niche, i.e., the span of conditions which the specie is exposed to.

The model fits the portion of the niche represented by the observed records.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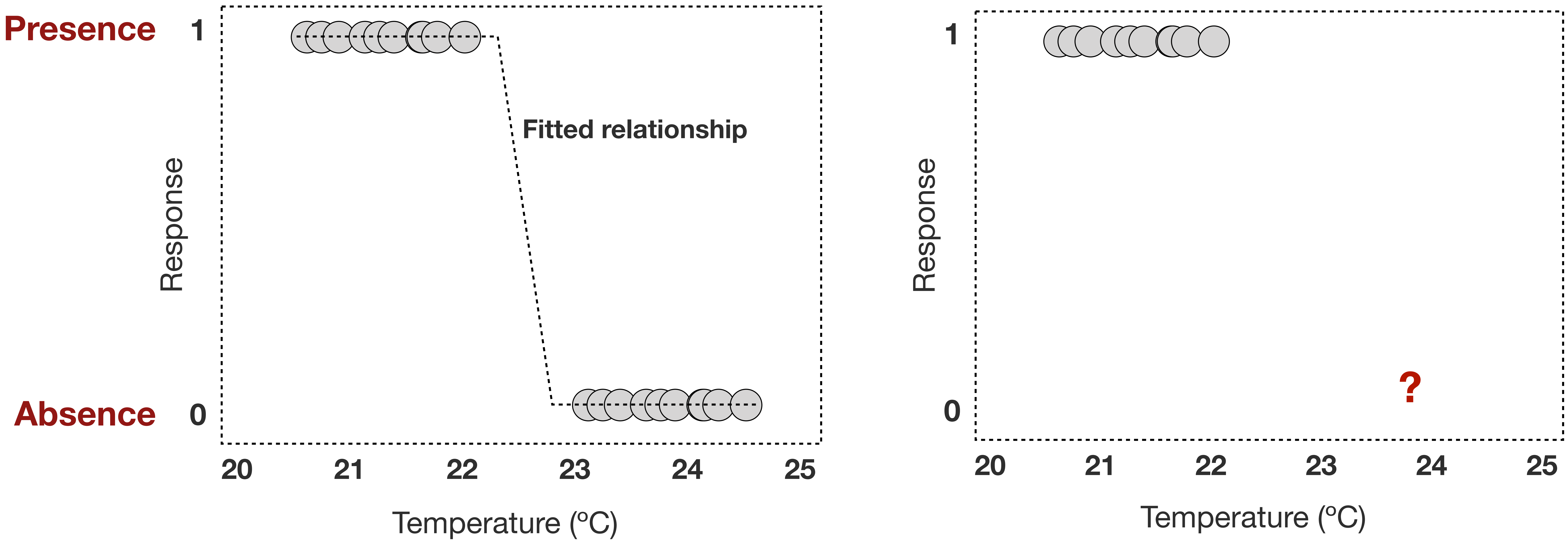
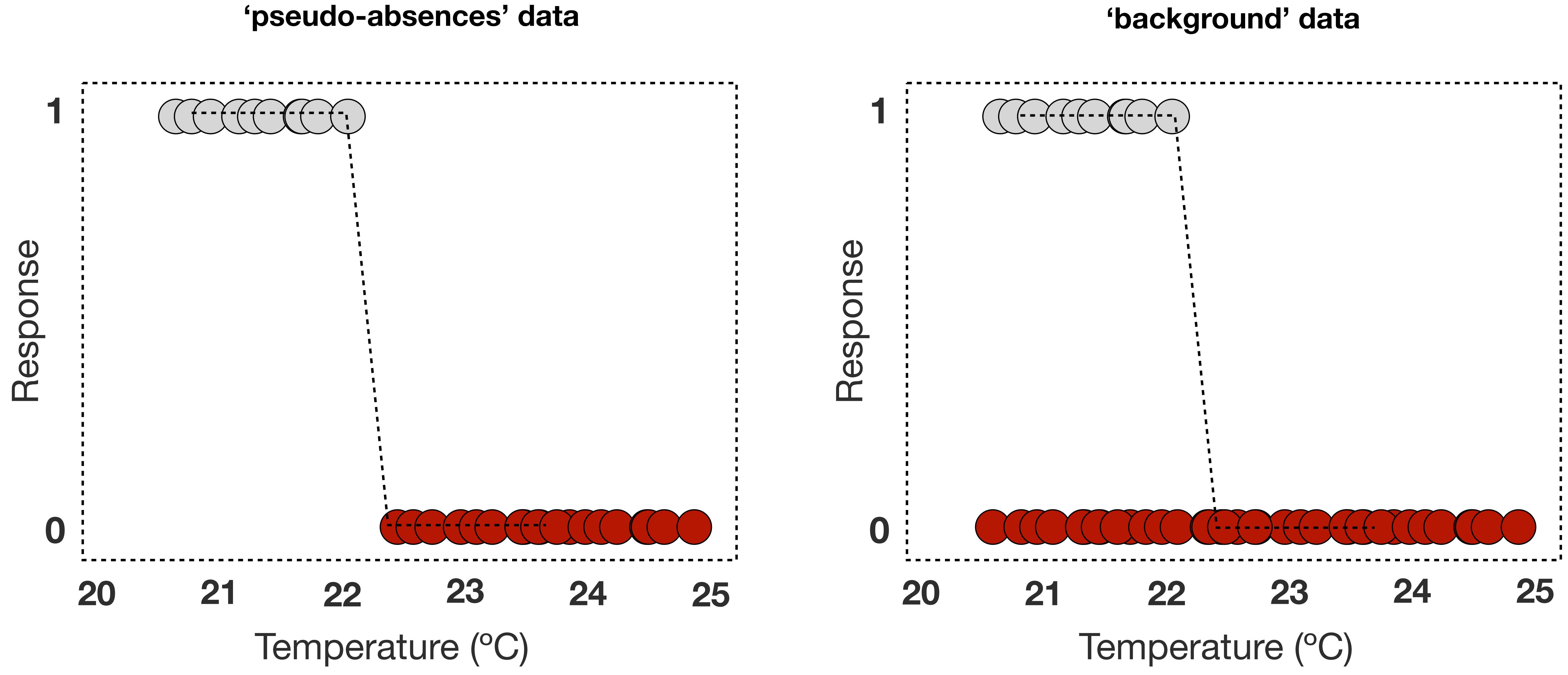


Handling Presence-Only Data

Correlative models infer **the relationship between species presence and absence records and the environment**, but **absences are often unavailable or unknown**, leading to **presence-only datasets** (common for marine species).



Strategies for presence-only modeling

Pseudo-absences: Artificially generated in sites of the study area where presences have not been recorded. Regression algorithms can be implemented (e.g., GLM and BRT).

Background data: Artificially generated from the entire study area. Focuses on comparing the conditions where the species occurs vs. the entire study area (e.g., MaxEnt).

Biased records

Detectability

The absence of a recorded observation does mean the absence of the species. A species **might be present but unobserved** due to cryptic behavior, daily/seasonal migrations, or simply a lack of survey effort.

Some species more easily detected. The **more mobile the species, the greater is the detection problem**.

Problem: Treating un-surveyed areas as "true absences" severely penalizes the process of capturing the niche, making presence-only techniques necessary.

Strategies for presence-only modeling

In global databases like GBIF or OBIS, true absences are almost never recorded.

Sensitivity and stabilization: When generating absences, it is crucial to understand that models respond differently to the different placement of these artificial points. Require multiple, replicated runs to stabilize predictions and ensure robust results.

How many absence records?

Pseudo-absences:

Linear models: a large number (typicality 10,000).

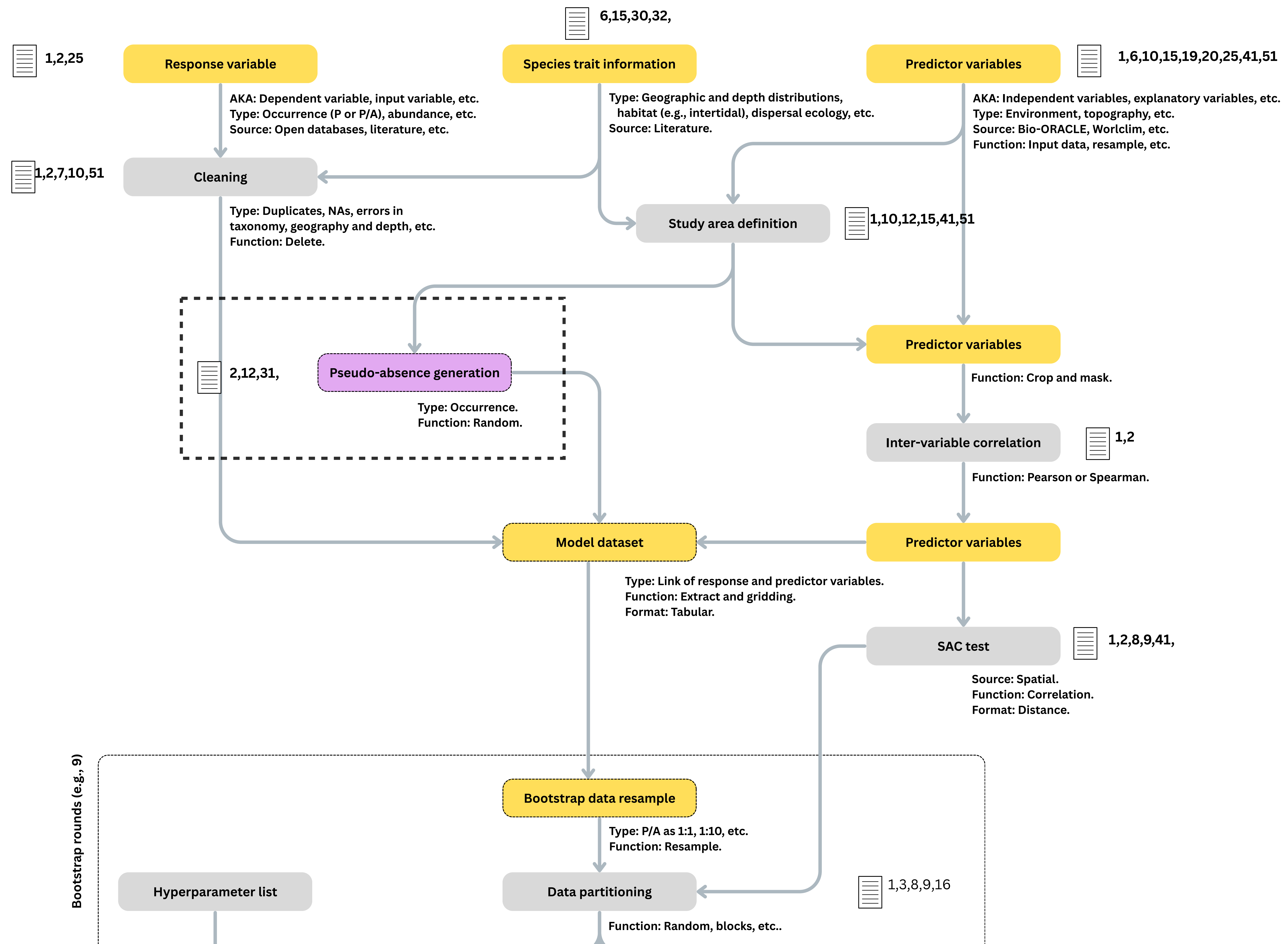
Advances machine learning models (e.g., BTR and XGBoost): the same number as presence records (when less than 1000 P, 10 runs with at least 100 PA).

Background data:

Maximum entropy (MaxEnt): a large number (typicality 10,000).

Barbet-Massin et al., 2012

biodiversityDS.



Environmental variables: selection

Ecological relevance is key:

Choose variables **known or hypothesized to directly influence the species' survival, growth, or reproduction**. Consider physiology and life history (literature).

Using large datasets is an approach with stronger criticism.

Important marine variables:

Physical: temperature, salinity, bathymetry, currents velocity, sea ice conditions.

Chemical: oxygen, nutrients (e.g., nitrate and phosphate).

Biological: chlorophyll-a (proxy for productivity).

Zurell et al., 2020

Environmental variables: selection

Marine mammals (e.g., whales, dolphins, seals):

Often driven by **temperature tolerances, prey distribution, and physical barriers** (e.g., sea ice).

Potential variables:

Ocean temperature: physiological tolerance limits.

Primary productivity / Chlorophyll: proxy for food web base supporting prey.

Bathymetry: depth limits for diving / foraging, habitat association.

Sea ice cover / sea ice thickness: Critical habitat for polar species (e.g., seals, polar bears).

Environmental variables: selection

Fish (e.g., Demersal Cod, Pelagic Tuna):

Temperature and oxygen are critical physiological drivers, plus factors related to food.

Potential variables:

Ocean temperature: physiological tolerance limits.

Salinity: physiological tolerance limits.

Dissolved molecular oxygen: physiological tolerance limit, especially in deeper or oxygen minimum zones.

Bathymetry: strong determinant of habitat zones (e.g., demersal species).

Primary productivity / Chlorophyll: proxy for food web base supporting prey.

Environmental variables: selection

Primary producers like macroalgae and seagrasses:

Requirements for light, nutrients, suitable substrate, temperature limits, water motion.

Potential variables:

Ocean temperature: physiological tolerance limits.

Salinity: physiological tolerance limits.

Photosynthetically Available Radiation (PAR): Essential for photosynthesis.

Diffuse attenuation: Measures light penetration depth, restricts photosynthesis.

Nitrate, Phosphate, …: Key nutrients for growth.

Environmental variables: selection

Corals (e.g., warm-water Reef Builders, cold-water Corals):

Temperature sensitivity, light (for symbiotic warm-water corals), substrate, water flow.

Potential variables:

Ocean temperature: physiological tolerance limits.

Salinity: physiological tolerance limits.

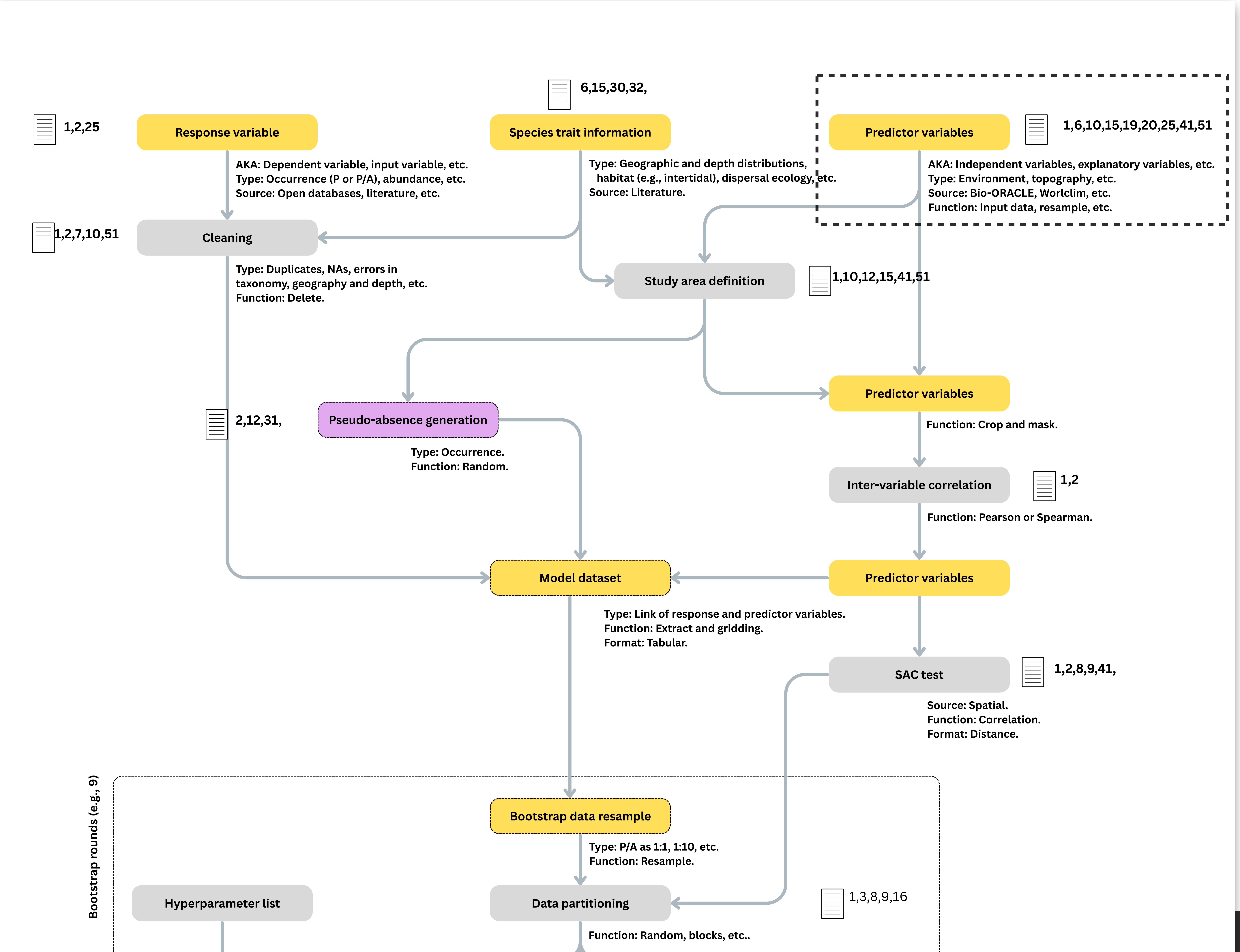
pH: affects calcification (building skeletons).

Dissolved molecular oxygen: physiological tolerance limits.

Bathymetry: defines depth range.

(For symbiotic zooxanthellate) PAR, Diffuse attenuation: Light for photosynthesis.

(For azooxanthellate) Primary productivity / Chlorophyll: Proxy for food supply.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Defining the study area

Calibration vs. projection area:

- **Calibration area:** where the model is trained (i.e., where the data is distributed).
- **Projection area:** where the model is applied to make predictions (can be broader, e.g., globally, future climates, new regions for invasion risk).

The **choice of calibration area significantly influences the model**, and therefore performance and transferability.

Defining the study area

Major question while defining the calibration area: Which areas has the species been able to access over time? (the species' accessible area, the 'M' in BAM).

The calibration area must encompass accessible areas to capture the range of conditions the species has experienced (suitable and unsuitable).

Rule of thumb: choose an area relevant to the species' dispersal ecology, large enough to capture the environmental gradients. Avoid environmentally truncated areas (e.g., sampling only the warm edge of a species).

e.g.,

The larvae dispersal potential of a coral species is 10 days; in the ocean this can be translated onto 200 km of maximum dispersal potential - the study area is defined as the area around any observed record using a buffer of 200km.

www.nature.com/scientificdata

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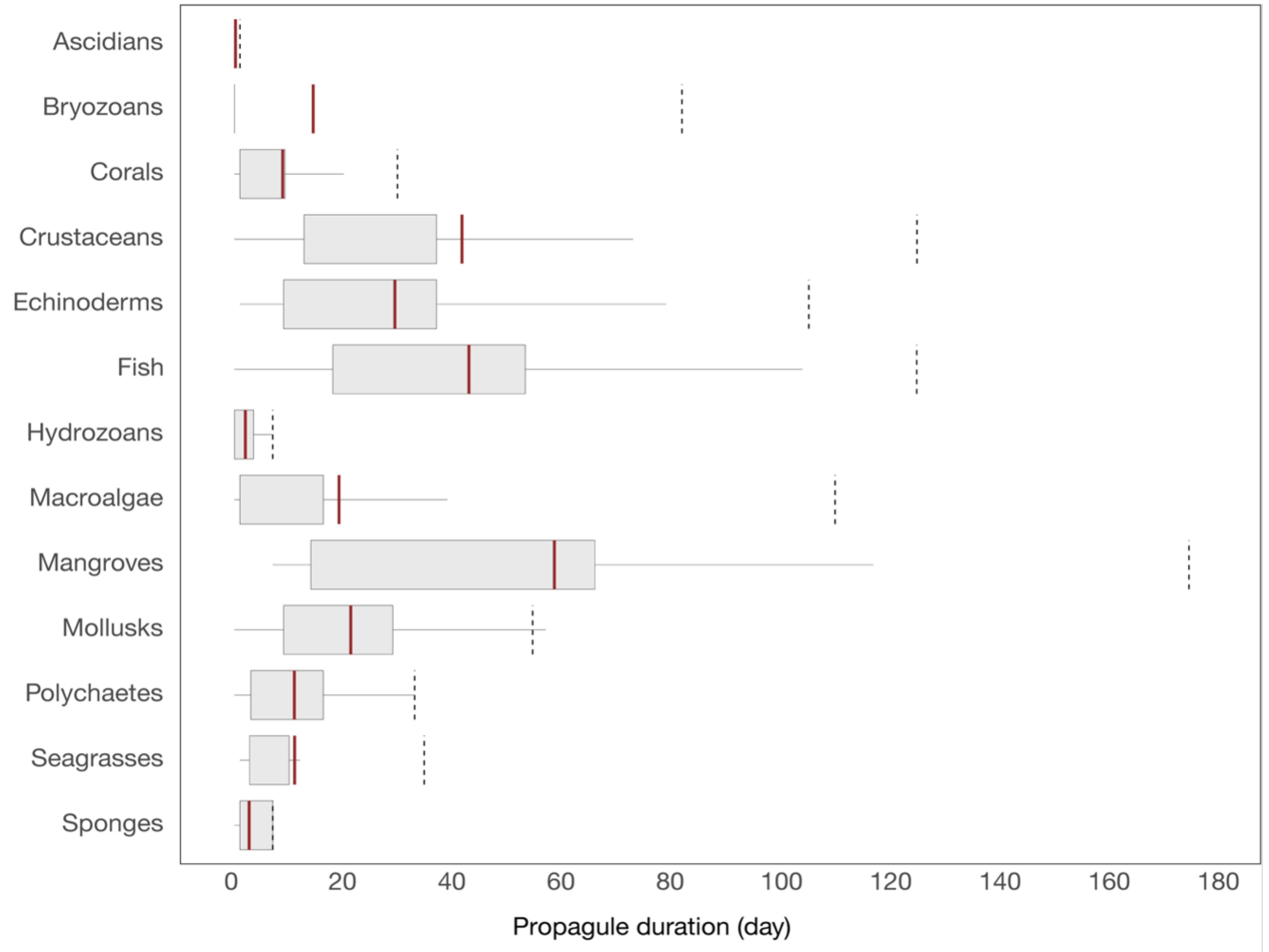
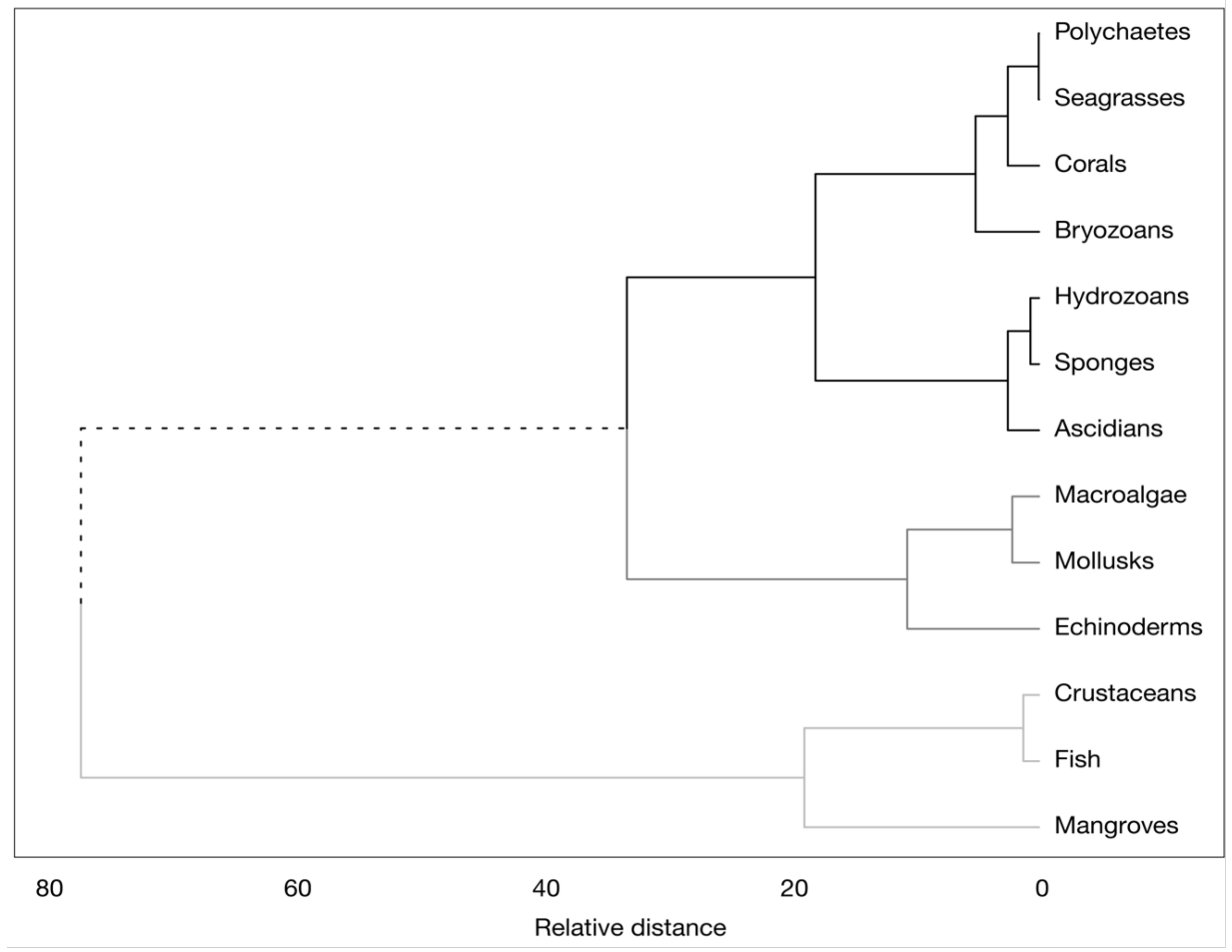
DATA DESCRIPTOR

Coastal oceanographic connectivity at the global scale: a dataset of pairwise probabilities and travel times derived from biophysical modeling

Jorge Assis[1,2] ✉, Eliza Fragkopoulou[1], Ester A. Serrão[1] & Miguel B. Araújo[3,4,5]

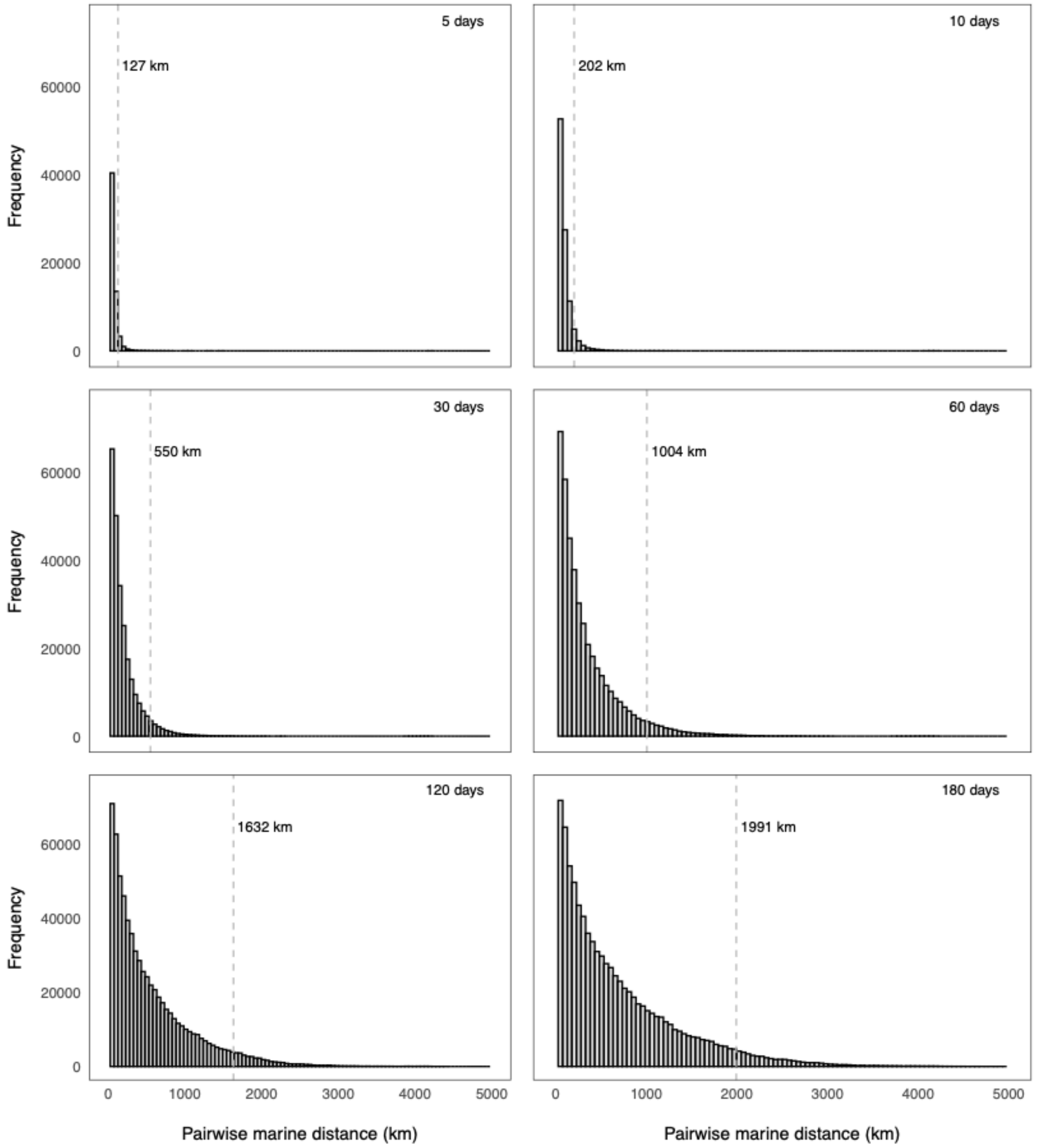
Ocean currents are fundamental drivers of marine biodiversity distribution, mediating the exchange of genetic material and individuals between populations. Their effect ranges from creating barriers that foster isolation to facilitating long-distance dispersal, which is crucial for species expansion and resilience in the face of climate change. Despite the significance of oceanographic connectivity, comprehensive global estimates remain elusive, hindering our understanding of species' dispersal ecology and limiting the development of effective conservation strategies. We present the first dataset of connectivity estimates (including probability of connectivity and travel time) along the world's coastlines. The dataset is derived from Lagrangian simulations of passive dispersal driven by 21 years of ocean current data and can be combined with species' biological traits, including seasonality and duration of planktonic dispersal stages. Alongside, we provide *coastalNet*, an R package designed to streamline access, analysis, and visualization of connectivity estimates. The dataset provides a new benchmark for research in oceanographic connectivity, enabling a deeper exploration of the complex dynamics of coastal marine ecosystems and informing more effective conservation strategies.



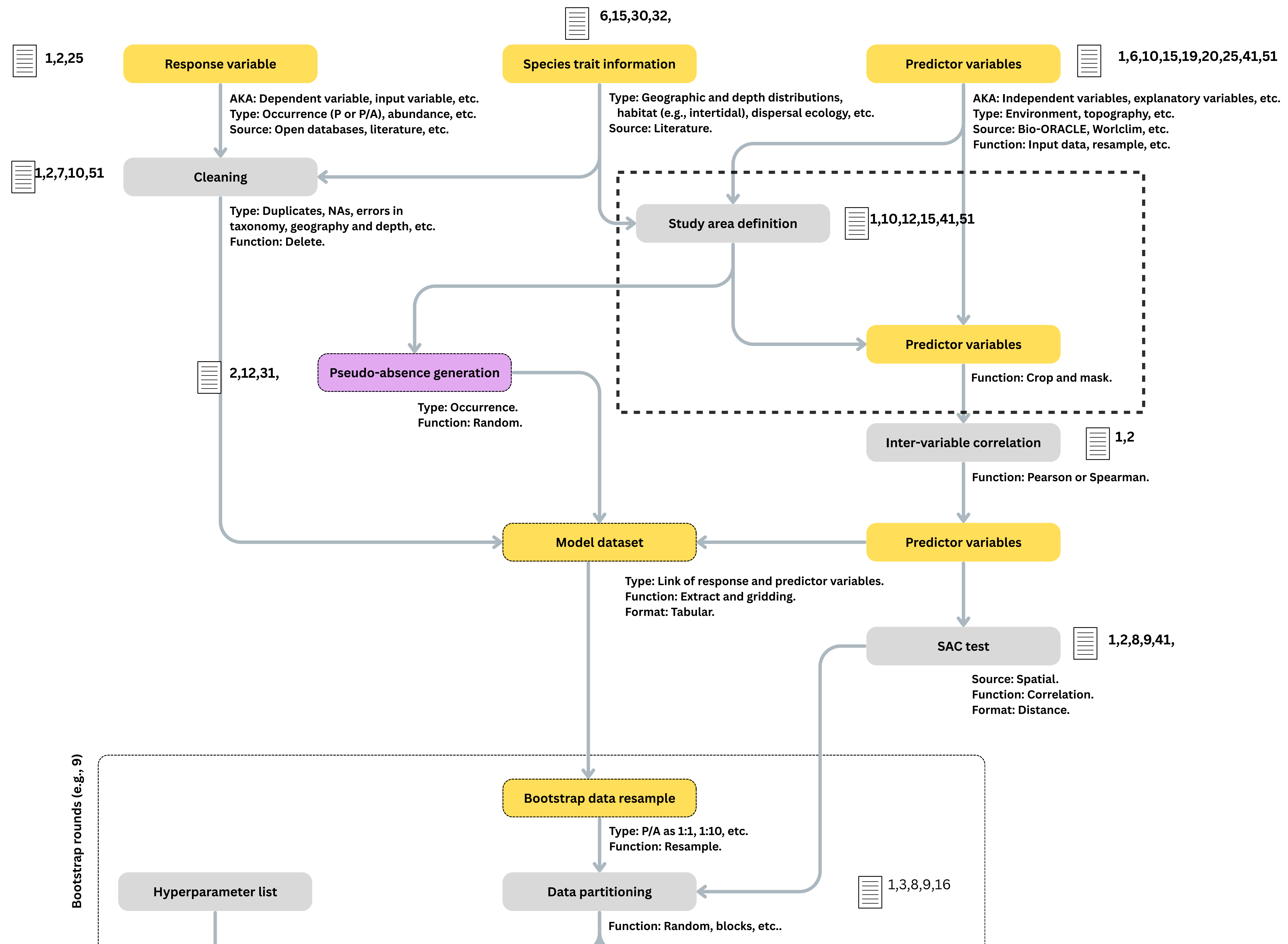


Distribution of propagule duration periods per ecological group. Vertical lines depict the mean (red) and the extreme 95th percentile (dashed) of propagule duration across observations. Dendrogram clustering groups based on propagule duration periods.

biodiversityDS.



Frequency of the distances of connectivity events for different propagule durations. Vertical dashed line depicts the 99th percentile of connectivity distances.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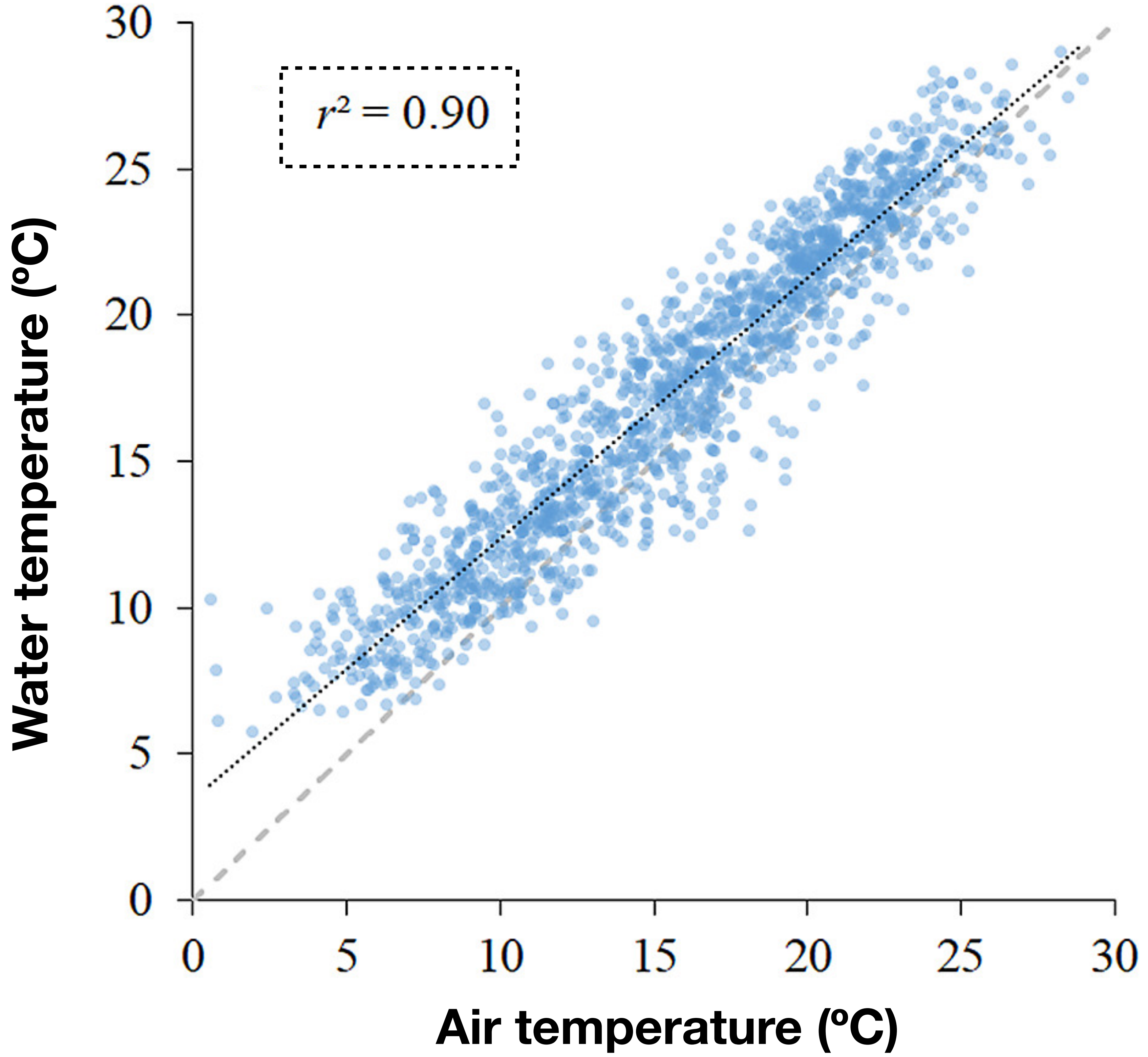


Multicollinearity: Many variables are highly correlated (e.g., Sea Temperature is often highly correlated with Air Temperature; Depth is correlated with Light Attenuation). Many algorithms (e.g., GLMs) have problems fitting correlated variables - impeding niche characterization. Models often fail when projected into future climates or new geographic regions, as the historical correlations might decouple under global change.

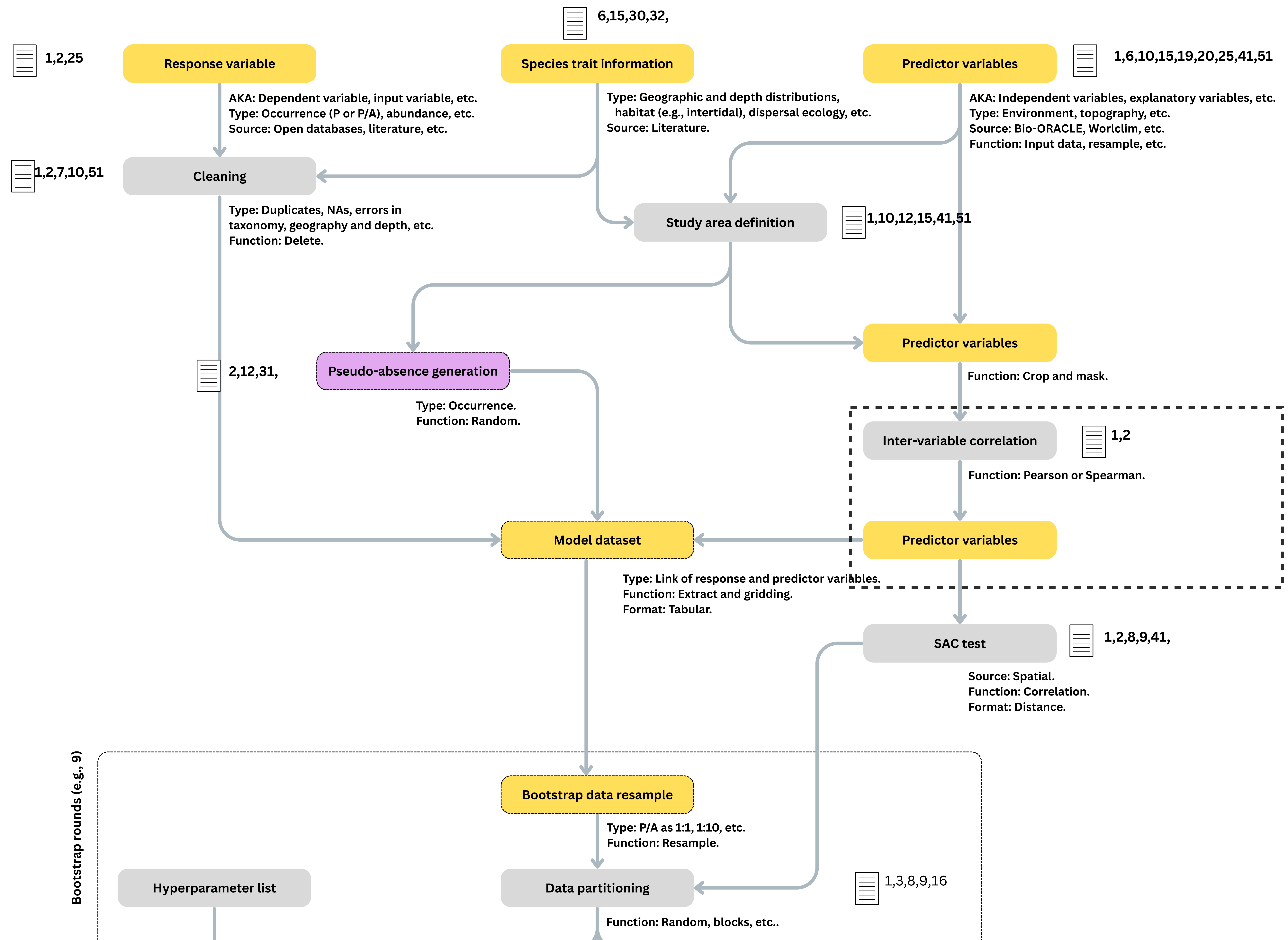
Elith et al. 2009

Correlation Index: The most common metrics used are Pearson's correlation coefficient. Range from -1 (perfect negative correlation) to +1 (perfect positive correlation). In ecological modeling, a common rule of thumb is to flag any pair of variables that $|r| > 0.75$.

Variance Inflation Factor (VIF): VIF can detect and quantify the severity of multicollinearity. A VIF of 1 indicates no correlation. A VIF between 1 and 5 is generally acceptable. A VIF > 10, indicates problematic collinearity.

Ecological Choice: If two variables are highly correlated (e.g., maximum temperature and average temperature), you cannot keep both in the model. The decision of which variable to drop should not be random - keep the variable that is biologically meaningful on the species' physiology or survival.

Elith et al. 2009


6,15,30,32,
1,2,25
Response variable
Species trait information
Predictor variables
1,6,10,15,19,20,25,41,51
AKA: Dependent variable, input variable, etc.
Type: Occurrence (P or P/A), abundance, etc.
Source: Open databases, literature, etc.
Type: Geographic and depth distributions, habitat (e.g., intertidal), dispersal ecology, etc.
Source: Literature.
AKA: Independent variables, explanatory variables, etc.
Type: Environment, topography, etc.
Source: Bio-ORACLE, Worlclim, etc.
Function: Input data, resample, etc.
1,2,7,10,51
Cleaning
Type: Duplicates, NAs, errors in taxonomy, geography and depth, etc.
Function: Delete.
Study area definition
1,10,12,15,41,51
Predictor variables
Function: Crop and mask.
2,12,31,
Pseudo-absence generation
Type: Occurrence.
Function: Random.
Inter-variable correlation
1,2
Function: Pearson or Spearman.
Model dataset
Predictor variables
Type: Link of response and predictor variables.
Function: Extract and gridding.
Format: Tabular.
SAC test
1,2,8,9,41,
Source: Spatial.
Function: Correlation.
Format: Distance.
Bootstrap rounds (e.g., 9)
Bootstrap data resample
Type: P/A as 1:1, 1:10, etc.
Function: Resample.
Hyperparameter list
Data partitioning
1,3,8,9,16
Function: Random, blocks, etc..

Model fitting

Correlative Algorithms: From GLMs to Machine Learning

The Core Objective: Construct a mathematical function that estimates how environmental variables (e.g., temperature, salinity, depth) constrain or promote a species' distribution.

The Machine Learning Revolution: Generalized Linear Models (GLMs) - highly interpretable and statistically grounded - can struggle to capture highly complex, non-linear ecological niches. Advanced Machine Learning (ML) techniques can learn such complex data.

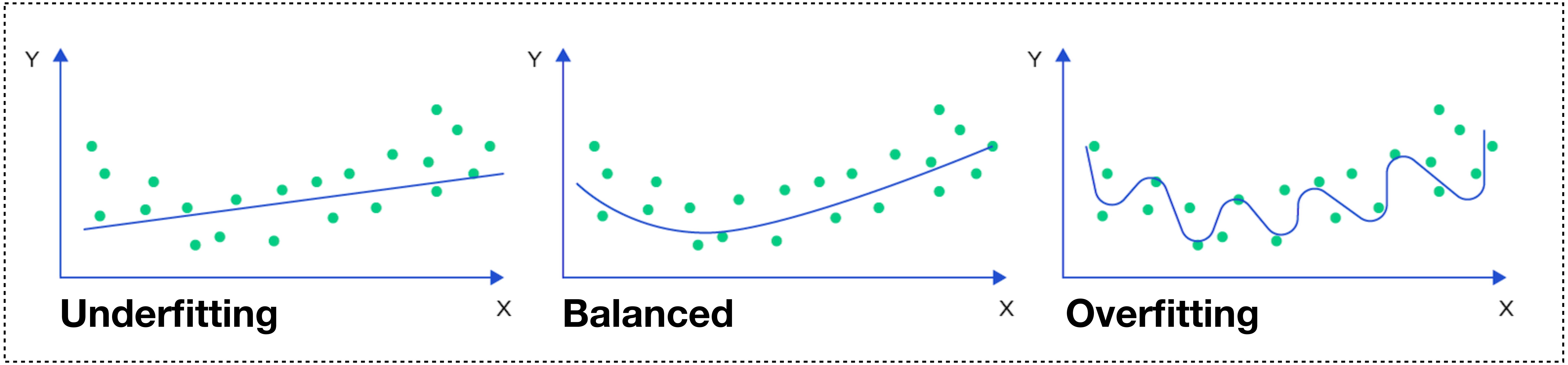
The Cutting Edge: We are currently witnessing an expansion into Deep Learning (Neural Networks) to handle massive spatial datasets. Furthermore, we are pushing beyond simple binary predictions (Presence/Absence) to spatially model continuous biological metrics, such as species Abundance and Biomass.

The Power of Machine Learning

Exceptional Predictive Performance: Algorithms such as Maximum Entropy (MaxEnt), Boosted Regression Trees (BRT), and Extreme Gradient Boosting (XGBoost) consistently yield extremely high predictive accuracy.

Can seamlessly integrate both continuous variables (e.g., temperature gradients) and categorical variables (e.g., substrate type).

Can automatically detect complex variable interactions - e.g., a species' tolerance to low oxygen severely decreases as water temperature increases (due to higher metabolic demand) - without the researcher having to manually flag interactions.

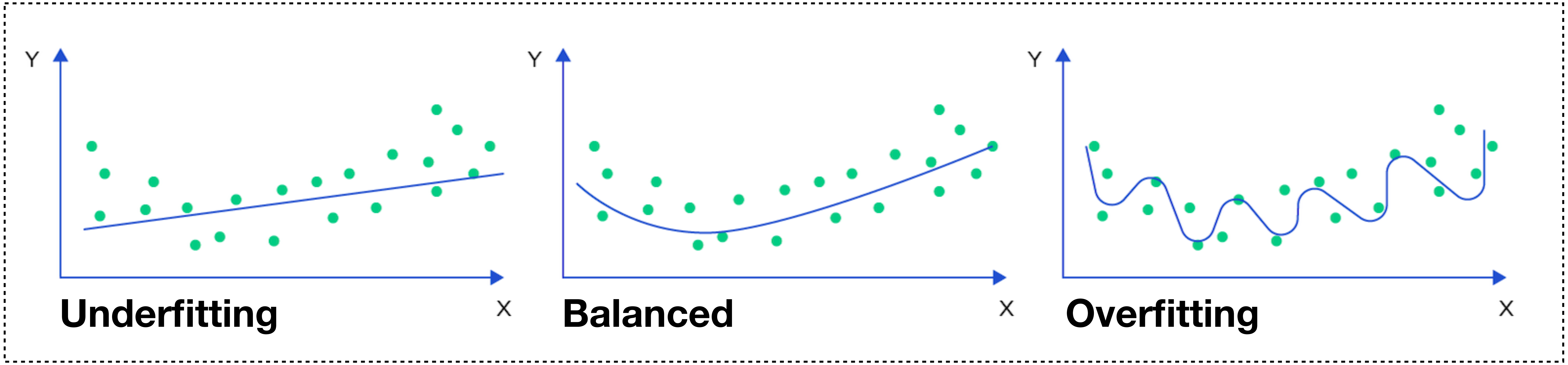


Model fitting pitfalls: underfitting

The model **is too simple to capture the underlying patterns in the data**.

Sources: simplistic algorithms (e.g., a GLM for complex relationships). Few or irrelevant variables. Insufficient occurrence data to represent the species' niche. Excessive regularization during model fitting.

Issues: Poor predictive performance. Species-environment relationships are missed, leading to unrealistic outputs.



Model pitfalls: overfitting

The **model learns the training data too well**, **including random noise and very specific patterns of the sample**, rather than the general patterns.

Sources: overly flexible model algorithms. Including too many environmental variables. Not applying appropriate regularization during model training.

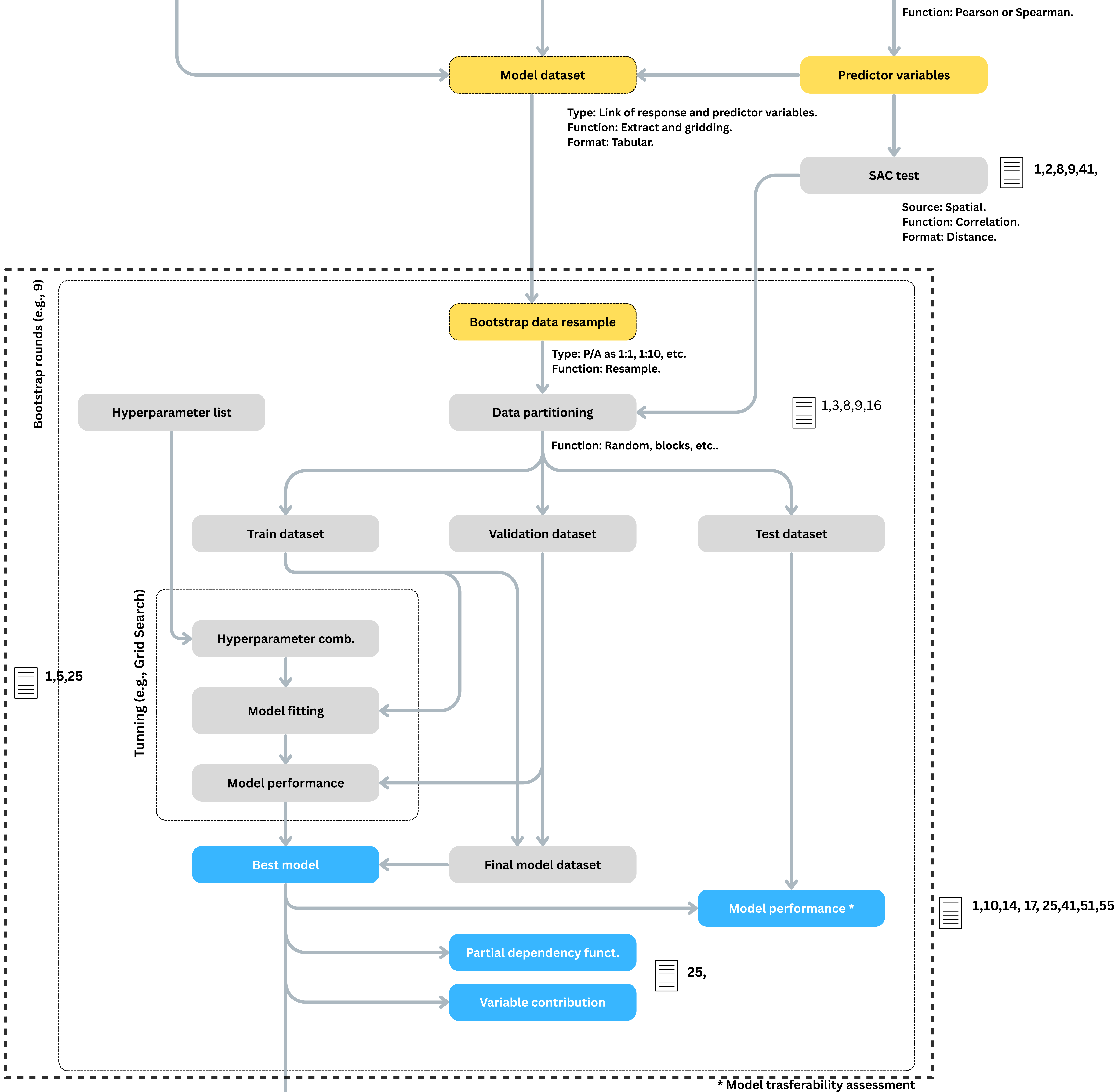
Issues: performs poorly on new data (e.g., future). Creates complex, biologically implausible distribution maps (e.g., distributions in tiny, isolated pockets based on noise).

Desired properties of model fitting

Deductive and general: to corroborate hypotheses around the causes of the pattern explained by the model, aiming at achieving ecological conclusions.

Simple: highlight a few mechanisms without becoming entangled in complex interactions between variables.

Parsimonious: preferring the simpler of two equally adequate models; favor simple explanations over complex ones.


Function: Pearson or Spearman.
Model dataset
Predictor variables
Type: Link of response and predictor variables.
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Format: Tabular.
SAC test
1,2,8,9,41,
Source: Spatial.
Function: Correlation.
Format: Distance.
Bootstrap rounds (e.g., 9)
Bootstrap data resample
Type: P/A as 1:1, 1:10, etc.
Function: Resample.
Hyperparameter list
Data partitioning
1,3,8,9,16
Function: Random, blocks, etc..
Train dataset
Validation dataset
Test dataset
Tunning (e.g., Grid Search)
Hyperparameter comb.
1,5,25
Model fitting
Model performance
Best model
Final model dataset
Model performance *
1,10,14, 17, 25,41,51,55
Partial dependency funct.
25,
Variable contribution
* Model trasferability assessment

The Danger of Algorithm Default Settings

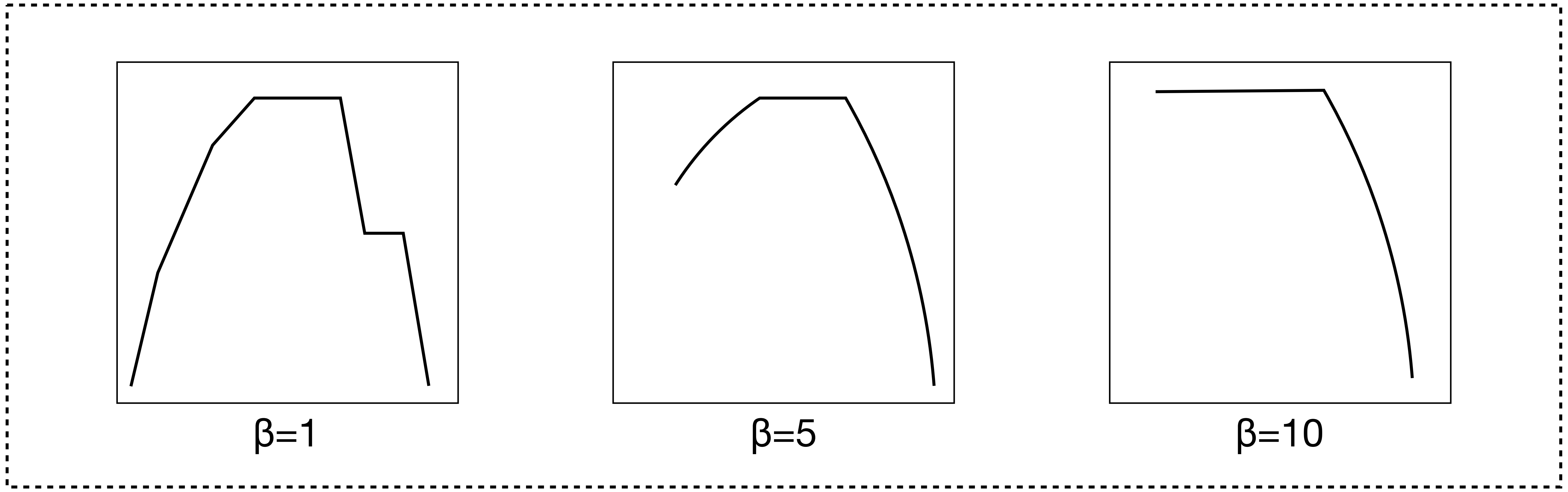
The "Black Box" Trap: Because ML algorithms are so powerful, they are highly susceptible to overfitting if left completely unrestricted.

Defaults are Dangerous: Relying on the software's "default settings", i.e., "default Hyperparameters" is a massive pitfall. There is no universal "rule of thumb" because every dataset context is unique.

The Solution: Hyperparameter Tuning - researchers must actively constrain the algorithms by tuning their hyperparameters. By testing a vast span of parameter values, we assess the model's ability to generalize, rather than just memorize.

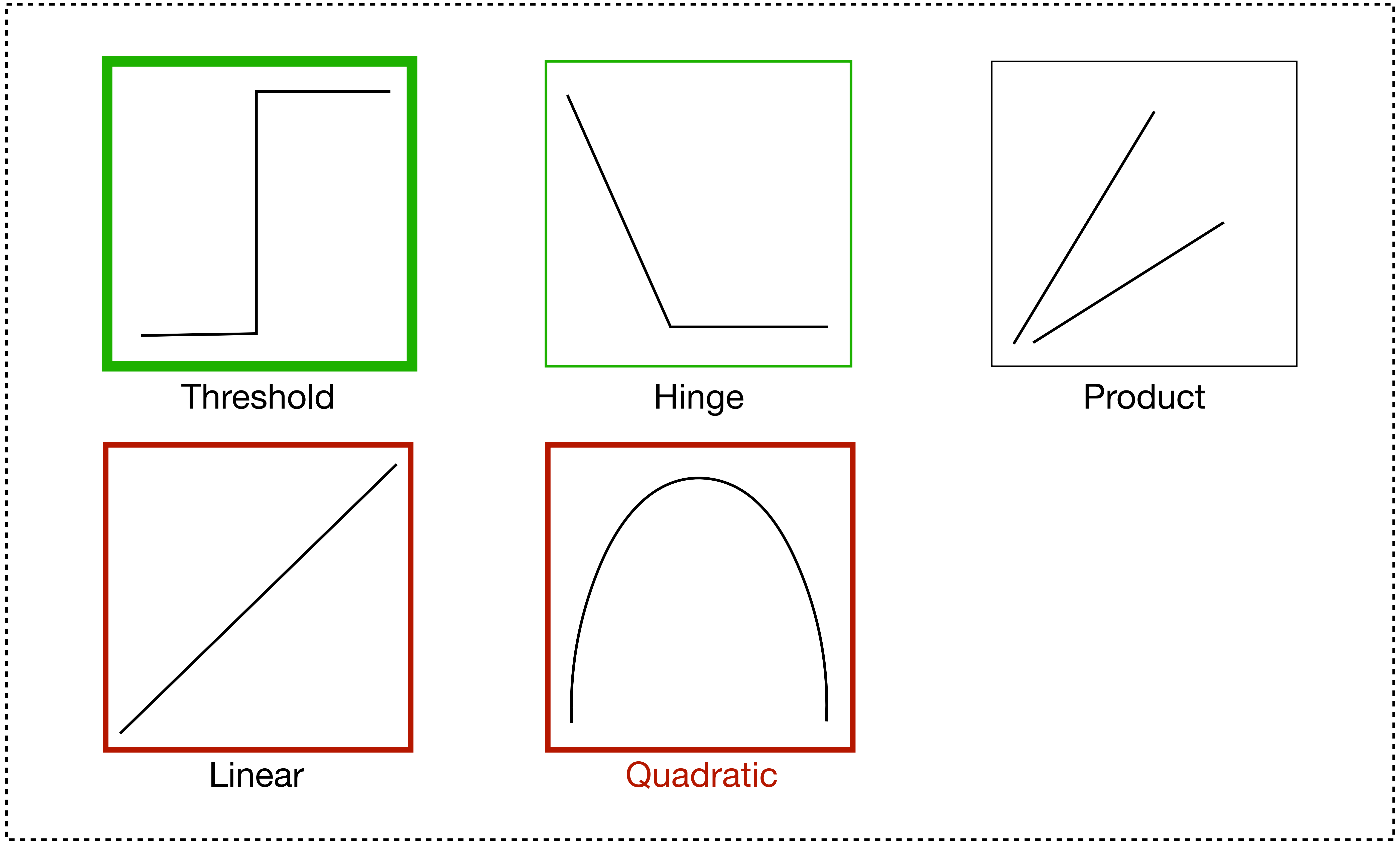
MaxEnt hyperparameters

A key parameter of **MaxEnt** is **regularization**, which controls **fit constraints and therefore strongly impacts overfitting**.



Higher regularization (β multiplier) values reduced the smoothing / relaxation of curves, which in term, can reduce overfitting.

To much regularization can lead to underfitting.

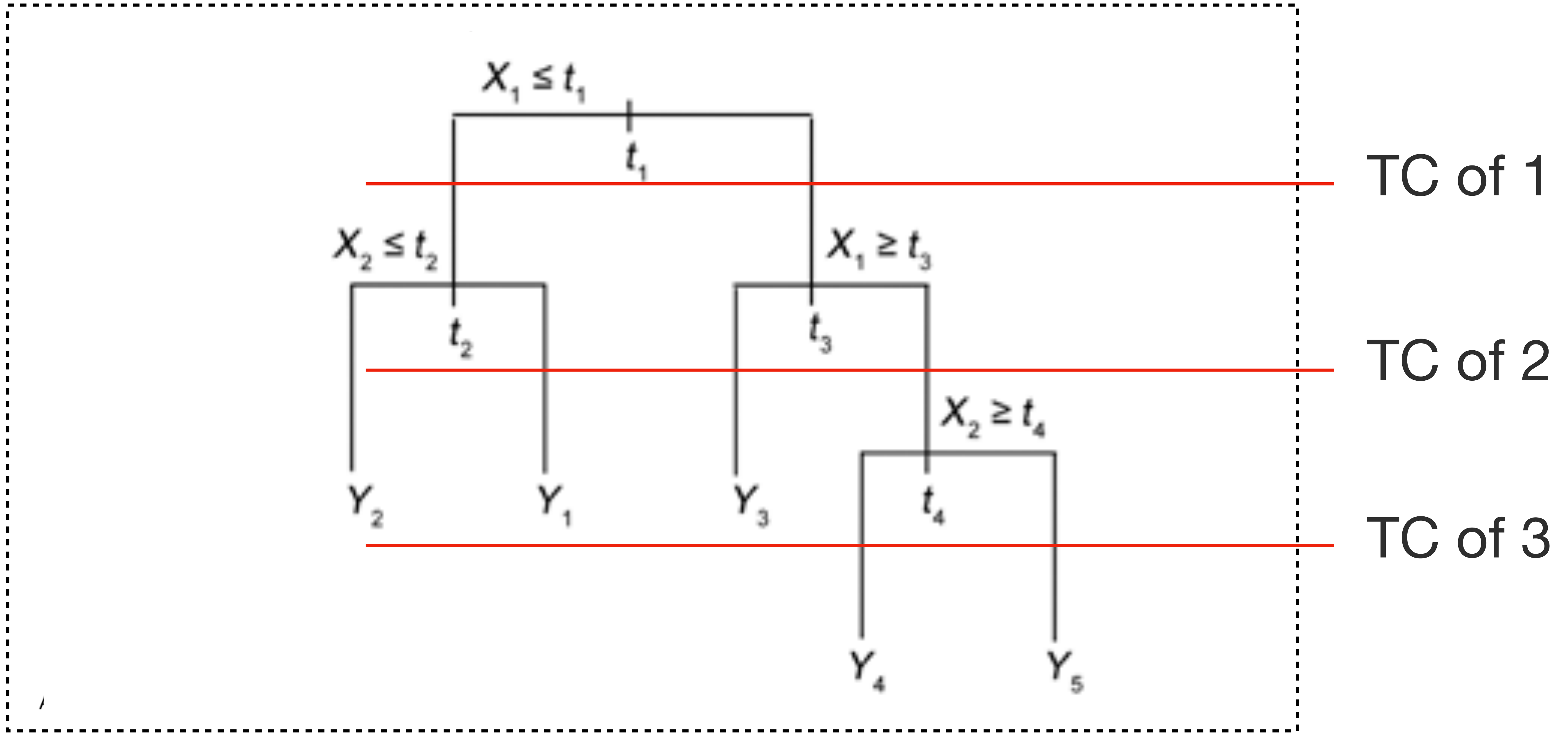


A key parameter of MaxEnt is **feature type - the type of the function**.

We tend to **exclude features that are more prone to lead to overfitting or to underfitting.**

Boosted Regression Trees hyperparameters

Tree complexity: a TC of 1 (single decision; two terminal nodes), a TC of 2 fits a model with up to two-way interactions, and so on. **TC controls complexity**, higher / lower can translate into overfitting / underfitting.

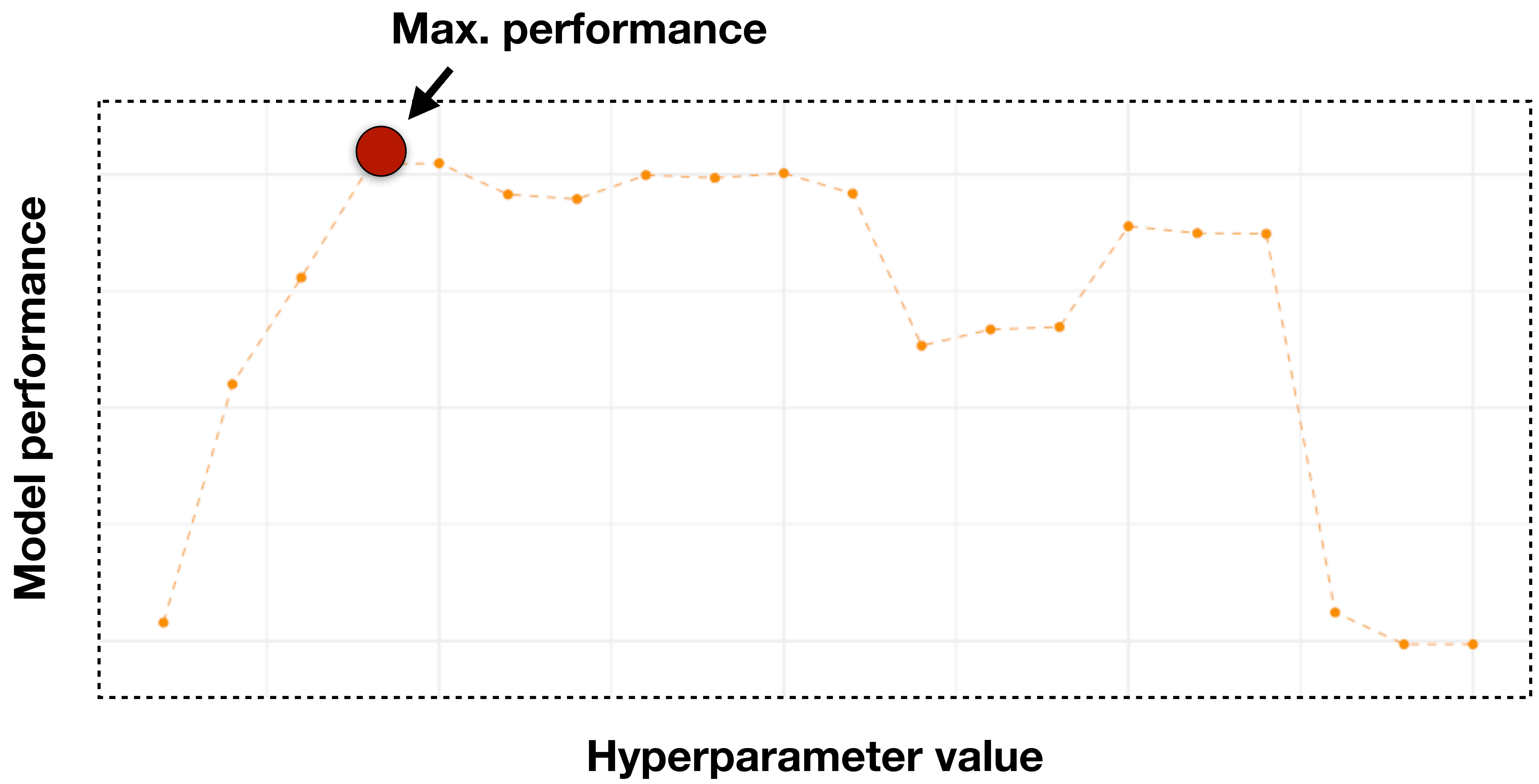


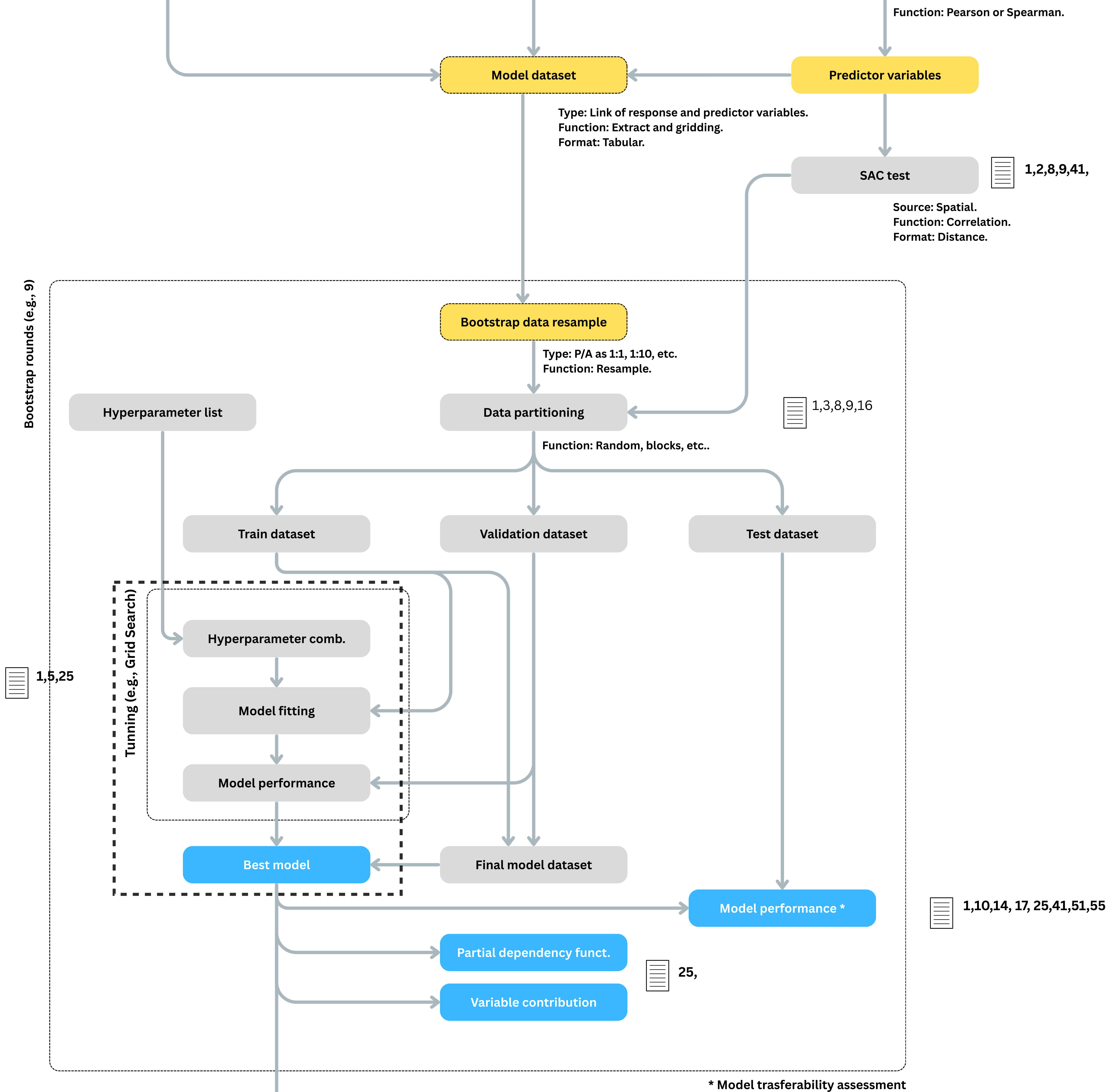
Hyperparameter optimization

How to tune these parameters to improve model fitting?

The approach relies on testing a span of parameter values and finding which value maximizes the performance of models.

Which parameter value has higher performance in validation data?





biodiversityDS.

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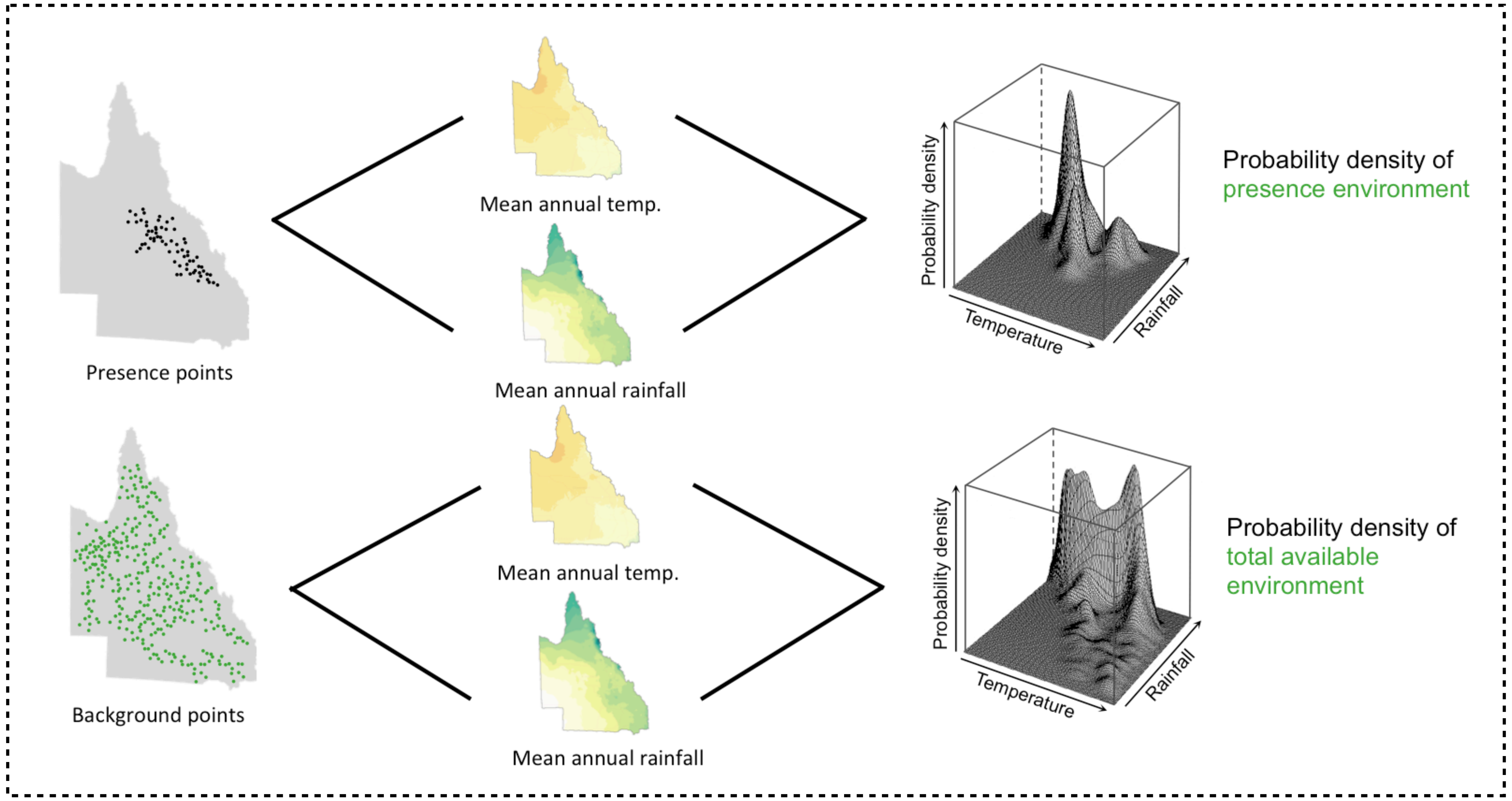
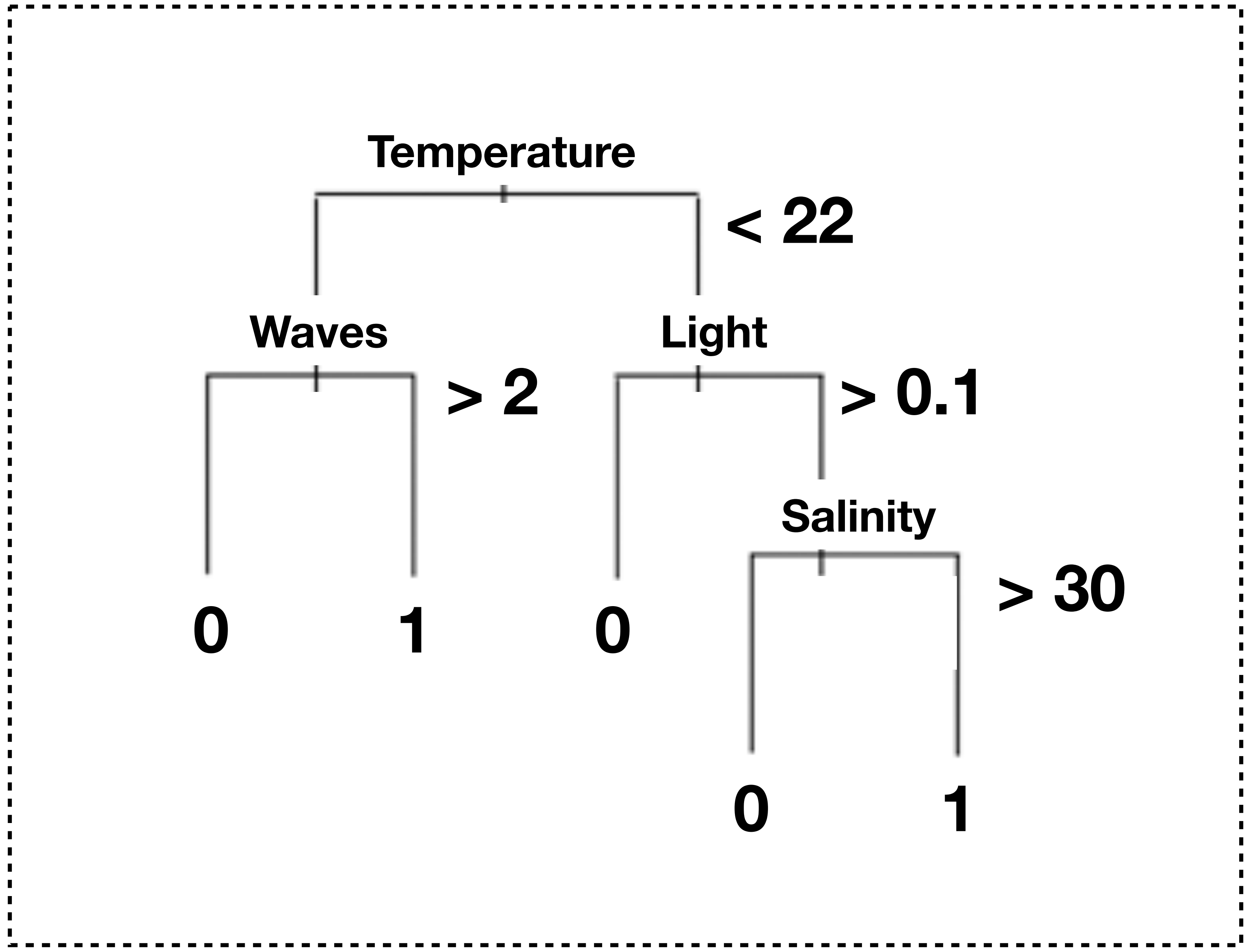
Assis et al., 2024: https://doi.org/10.1038/s41477-024-01735-7



Maximum Entropy fitting made by **comparing the density in the environmental conditions at the locations where the species has been found, with the environmental conditions across the study region.**

MaxEnt **outputs the environmental suitability of a species in the study area** (i.e., the potential distribution).



Boosted Regression Trees is a flexible and high performance algorithm, which combine the strengths of **decision trees** (models that relate a response to their predictors by recursive binary splits) **with boosting** (combines simple models to give improved predictive performance).

BRT gives the probability of a species occurring in the study area.

Machine learning algorithms

Construct a function that estimates the effect of environmental variables on the distribution of a species.

e.g., Maximum Entropy; Boosted Regression Trees; Extreme Gradient Boosting.

Advantages: (1) high predictive performance; (2) can use continuous and categorical variables; (2) automatically detect interactions between variables**; (3) **includes specific parameters that protect against overfitting**;

Major limitations: susceptible to overfitting if not properly parameterized.

*** when a variable has a different effect on the outcome depending on the values of another variable;*

e.g., species' tolerance to oxygen often depends on temperature; increased metabolic oxygen demand - higher temperature, higher oxygen demand.